

Highlights

Evaluation geometry and the limits of cross-region transfer in pre-fire thermal wildfire prediction

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- Scoring the same model at the fire, not region-wide, costs 0.143 AUC
- The cause is the negative pool, not prevalence: +0.155 against -0.000
- Thermal predictors add +0.06 to +0.15 AUC within all five fire regions
- Equalising the frame lifts transfer 0.540 to 0.617, six below chance to one
- On comparable evaluation frames, none of 20 diagnostics orders transfer

Evaluation geometry and the limits of cross-region transfer in pre-fire thermal wildfire prediction

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Abstract

Pre-fire thermal dryness separates a fire year from a normal year, but models built on it are rarely tested outside the region where they were fitted. Five Mediterranean wildfire regions were analysed on about 500 m cells, with MCD64A1 labels and spatially blocked validation, adding six thermal predictors to a terrain, fuel and greenness baseline.

Local skill does not travel, and most of it is lost before the region changes. Within-region ROC-AUC rose by +0.056 to +0.153 under blocked cross-validation, but by +0.022 [−0.032, +0.077] when a whole burn scar is withheld, and by +0.004 [−0.028, +0.036] across twenty transfer directions, with a sign that changes from pair to pair. The static baseline transferred no better, 0.537 against 0.541, so this is not a property of the dynamic block. On **identical cells**, a model holding the held-out scar scores 0.634, one without it 0.552 and one fitted 306 to 2,802 km away 0.555, so neither withholding the fire nor crossing the region is shown to cost anything. What is measurable is the evaluation area itself, worth **0.143** [+0.077, +0.208] of the fall from a region-wide 0.776, because its negatives are all fire-adjacent and therefore

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the hardest in the region.

The failure cannot be anticipated. Of twenty candidate diagnostics only two had intervals excluding zero, both measuring sign agreement over a selected feature subset ($\rho = 0.84$, post-selection). Recomputed on the equalised frame that statistic is unanimous and has no variance left, and its continuous counterpart falls from $\rho = 0.50$ to 0.12 , so no diagnostic tested here was shown to order transfer once the frames are comparable.

Most of the apparent mechanism is an artefact of the evaluation frame. Five predictors reverse direction between regions on the frames as drawn, but those frames enclose very different far fields, from 2 % to 63 % of cells beyond 10 km of any fire, on higher ground. A 10 km collar, which drops no burned cells, makes all five agree in sign on elevation, LST and TVDI, lifts transfer from 0.540 to 0.617, drops the directions below chance from six to one, and leaves no reversal bootstrap-supported under our own criterion. The agreed sign is not the one the dryness framing predicts: hotter surfaces burned less everywhere, surviving stratification within elevation and greenness.

Label-free alignment pushed fourteen of twenty directions towards chance rather than repairing them. Transfer skill has to be measured, not inferred from similarity.

Keywords: wildfire, model transferability, area of applicability, land surface temperature, domain adaptation, concept shift

1. Introduction

Wildfire is a defining disturbance of Mediterranean-basin landscapes, and a changing climate is reshaping where and how it burns [29]. The pressure to anticipate *where* fire will occur has produced a large and still-growing literature on fire susceptibility mapping. It has recently been synthesised from both a methodological and an application perspective [42, 14]. The dominant pattern is well established. Geospatial predictors are assembled over a study region and paired with a historical record of burned area. A supervised classifier is then fitted, most often a random forest [5]. The resulting susceptibility surface is published with a cross-validated AUC in the 0.85 to 0.95 range.

This paper is about a property that the pattern does not report. Predictors that describe the *state* of a surface in a particular season are more informative about that season than predictors that describe a place. They are also, as we show, not portable, and a large part of this paper is about how much harder that is to establish than it looks: the reading we first reached, that the relationship is reparameterised locally and points in different directions in different places, does not survive a correction to how the study regions were evaluated (Section 4.10). The dynamic block is worth +0.056 to +0.153 ROC-AUC inside every region under blocked cross-validation, +0.022 when a whole burn scar is withheld, and contributes +0.004 between them, an estimate whose interval spans zero, with a sign that is a property of the source-target pair rather than of the block.

The obvious reading of that pair of numbers is a trade-off, and our own earlier framing took it. The data do not support it. The static baseline trans-

fers at a mean of 0.537 against the thermal model’s 0.541, so the predictor class that ought to travel does not travel either, and removing the reversing predictors returns +0.014 of transfer on an interval that also spans zero. Two nulls on the portability axis are not an exchange. What is measured is a local gain that does not export, and a local cost when the reversing predictors are removed, with no compensating transfer gain on either arm.

1.1. Portability goes unmeasured

A susceptibility model’s reported skill is almost always an estimate of *within-region* performance. Held-out folds come from the same study area and season, often from the same fire event, and where folds are drawn at random over cells, spatial autocorrelation inflates the estimate further. The problem is documented across ecological modelling [33, 31] and is addressed by spatially blocked cross-validation [39, 25]. Blocking removes the inflation that random folds produce, and it is the right correction for what it corrects. It does not make the estimate honest about a fire the model has not seen. Section 4.3 measures that on identical cells: a model with the held-out burn scar in its training data scores 0.634 on that scar’s area, and withholding the scar gives 0.552. Part of the apparent gap between a region-level figure and either of those is the evaluation area rather than the model.

Because only that side of the ledger is reported, portability is never entered at all. A predictor block is adopted on the strength of the increment it delivers inside its training footprint, and whether that increment survives a change of region is not asked, even though any regional product built from locally trained models implicitly promises generalisation beyond that footprint. For fire specifically the transfer question is asked far less often

than in adjacent fields. Meteorological fire-danger indices are known not to port cleanly between fire environments [32], and the two recent studies that test model transfer systematically, across Mediterranean countries [8] and US counties [16], both report that it largely succeeds between similar
55 regions. Both transfer models whose dominant predictors are *spatially stationary*: terrain, human modification, night-time lights, population density, long-term climatologies. Each describes a place rather than a season, which is the class our thesis expects to travel well. Species distribution modelling has examined transferability far more systematically [46], and there geographic
60 and environmental similarity do *not* reliably predict transfer success [41, 34].

1.2. Pre-fire thermal dryness is the natural test case

The dynamic class used here is pre-fire thermal dryness, and it is the class most plausibly *expected* to transfer. Standard susceptibility predictors are terrain, fuel type, greenness, long-term climatology and proximity to
65 roads and settlements, and most are static or near-static over the timescale at which fire danger varies. Such a set explains poorly why a particular summer burned and the preceding one did not. What changes between those summers is the state of the surface: fuel dryness, moisture stress, and how anomalously hot the land surface has become relative to its own baseline.
70 Satellite thermal observation gives direct access to part of that state through fuel moisture content [47], and pairing surface temperature with a vegetation index was established as a live-fuel-moisture estimator for fire-danger rating by Chuvieco et al. [7]. The Temperature-Vegetation Dryness Index [35] formalises that feature space into an internally normalised measure. In
75 principle it should be less exposed to absolute-temperature offsets between

regions than raw land surface temperature. That theoretical portability advantage is tested here, and it is not found: the two internally normalised channels transfer no better than the four absolute ones, at means of 0.544 and 0.548 over the twenty directions (Appendix A(f)).

80 That pre-fire thermal state carries genuine information about subsequent fire is established [19, 21, 20, 9]. What has not been established is whether the predictive *skill* of a classifier trained on it in one fire region survives strict, label-free application to another. The physics linking moisture stress to combustion is universal, so portability should be most expected for this
85 class, which is what makes it diagnostic: a loss here cannot be dismissed as a peculiarity of a locally defined covariate. That expectation is the motivation for the design, not a finding of it. Section 4.10 reports that the signed associations run the other way, a hotter pre-fire surface being associated with less burning in all five regions and surviving stratification within elevation and
90 greenness, so on this cohort the block behaves as a proxy for fuel availability rather than as a moisture-stress index. We keep the framing because it is why these predictors were chosen and why their failure to travel is informative, and we state the contradiction where the evidence appears rather than adjusting the motivation after the fact.

95 *1.3. Why the loss is invisible to the diagnostics in use*

A deficit on transfer can arise two ways. Under **covariate shift** the marginal distribution of the predictors differs while the predictor-response relationship is preserved, which is in principle correctable without target labels, by per-region standardisation or covariance alignment such as CORAL
100 [37], an approach with an established remote-sensing literature [38]. Under

concept shift the conditional relationship itself changes: a predictor positively associated with burning in one region is negatively associated with it in another. No label-free realignment of input distributions can repair a sign reversal, because the information needed to detect it is exactly the information being withheld. The taxonomy is canonical [27].

This distinction is what makes the failure hard to anticipate. The diagnostics the field relies on to decide whether a model may be applied to new ground, above all area-of-applicability measures built on dissimilarity in predictor space [23, 24, 18], are marginal by construction. A target region can sit well inside the training data’s predictor envelope while the relationship between those predictors and burning points the other way. This paper measures both sides on the same pairs.

1.4. Contributions

Three findings carry this paper. Each is stated at the strength its interval supports, and the nearest prior work is named inside the claim.

Contribution 1. Where a model is scored decides what it appears to know, and the effect is large enough to dissolve one of our own findings. That evaluation extent inflates AUC is established in species distribution modelling, by Lobo et al. [17], VanDerWal et al. [40] and Barve et al. [3] (Section 2.5); what is new here is a magnitude, under a controlled design that earlier work did not run, and its consequences for a live wildfire result. Holding the model, the predictors and the fitting fixed and changing only which cells are scored, moving from the whole region to the burn scar and its 2 km collar costs **0.143 ROC-AUC [+0.077, +0.208]**, from 0.776 to 0.634. That exceeds the predictor-block increments this literature pub-

lishes as findings, so it is not a second-order correction. A control isolates the cause: matching a region sample to the scar area’s prevalence changes nothing, at -0.000 $[-0.003, +0.002]$, while scoring the same predictions on the scar area costs $+0.155$ $[+0.093, +0.217]$. It is the composition of the
130 negative pool, whose cells are all fire-adjacent and therefore the hardest in the region, not class balance.

Applied between regions, the same effect withdraws three of our own claims (Section 4.10). The five study areas enclose very different far fields, from 2 % of Montiferru’s cells lying beyond 10 km of any fire to 63 % of Bejís’s,
135 and that far field is higher ground. Restricting every region to a 10 km collar, **which drops no burned cells**, makes all five agree in sign on elevation, on LST and on TVDI; both bootstrap-supported elevation reversals disappear; mean transfer rises from 0.540 to 0.617; and the directions below chance fall from six to one. The sign reversal we had offered as the mechanism
140 of the residual is therefore mostly a property of how five rectangles were drawn. Under this paper’s own criterion **no reversal remains bootstrap-supported** in the equalised frame; the closest is the LST anomaly between Bejís and Evia, a point reversal whose Evia interval includes 0.5 by 0.003. The sign the five regions now agree on is also not the one the dryness framing
145 predicts: hotter pre-fire surfaces burned less in every region, and that survives stratification within elevation and greenness deciles, so on this cohort the block behaves as a proxy for fuel availability rather than as a moisture-stress index (Section 4.10). A reversal also appears between two fires eleven months apart inside one study area, where the frame is fixed by construction
150 (Section 4.9), but on elevation alone, on eleven positive-carrying blocks, and

with one fire per arm it is as consistent with event-to-event variation as with any regional property. The practical consequence is a reporting standard, stated in Section 5.8.

Contribution 2. Local skill does not travel, and correcting the
155 **frame does not rescue it.** The thermal block is worth +0.056 to +0.153 ROC-AUC inside every one of five regions under blocked cross-validation, with every bootstrap interval above zero and the result holding in both analysis populations. Much of that is a property of interleaved holdout: withholding a whole burn scar leaves +0.022 $[-0.032, +0.077]$. Across twenty
160 ordered transfer directions it contributes +0.004 $[-0.028, +0.036]$, an interval spanning zero under all four between-direction resampling units we computed, with a sign that varies by pair — paired deltas run from -0.148 to $+0.133$, twelve positive and eight negative, and dropping Evia alone reverses their mean. Two controls bound the reading. The static baseline
165 transfers at 0.537 against the thermal model’s 0.541, so the portability failure is not the dynamic block’s peculiarity. And equalised transfer of 0.617 against within-region skill of about 0.87 leaves most of the gap intact, so the frame correction of Contribution 1 narrows the mechanism without rescuing the result. The within-region half is not novel, comparable results existing
170 for these landscapes [1, 13]; the paired contrast against portability is. Dimarco et al. [8] transfer a predominantly stationary predictor set successfully across a comparable Mediterranean design, though their response variable is ignition rather than burned area (Section 5.7).

Contribution 3. The shortfall is invisible to every diagnostic
175 **that could be run before deployment.** Twenty candidate diagnos-

tics from five families are evaluated against observed transfer under one bootstrap framework. Area-of-applicability-style predictor-space dissimilarity [23, 24, 18], climatic and geographic distance, learned domain separability, niche-overlap statistics and burn-pattern regime distances were none of them shown to order the matrix; the domain classifier is at ceiling, with AUC $\geq$ 0.96 for every pair. Only two diagnostics have intervals excluding zero, and both measure conditional direction agreement, that is, whether each predictor’s signed association points the same way in both regions (Spearman $\rho = +0.84$ and $+0.81$ over sixteen directions from eight pairs). Both are computed on a data-selected subset of predictors, and their all-nine-feature counterparts span zero. **Contribution 1 then removes even that exception.** The sign-agreement statistic is built from the signed associations that the frame test shows to be artefacts, and recomputed on an equalised frame it is unanimous — 1.0 in every direction, with no variance left to correlate — while the continuous alternative falls from $\rho = +0.50$ to $+0.12$ (Section 4.10). The honest finding is therefore stronger and worse for the practitioner than it first appeared: **no diagnostic tested here, marginal or conditional, was shown to order transfer once the frames are comparable**, and the one that appeared to was reading how the study rectangles were drawn. The same point survives without any ranking, at the point estimates: the pair with the highest burned-niche overlap fails in both directions while the lowest transfers in both. That contradicts an expectation the fire literature carries [8, 16] and agrees with species distribution modelling [41, 34].

Two consequences follow, reported as supporting results rather than leading ones. Label-free alignment by standardisation and covariance alignment

[37] does not repair transfer; it compresses fourteen of twenty directions towards chance, and we believe this is the first application of covariance alignment to fire susceptibility. And because the residual is conditional, the resource that closes it is target labels: thirty-two labelled 5 km blocks recover
205 85 to 89 % of the target’s matched ceiling in three of six directions, 51 to 57 % in two more, and 30 % in the sixth.

Alongside these, a leakage-audited, spatially blocked evaluation and transfer protocol is released with code, configuration and frozen outputs, so that most of this result can be re-run rather than taken on trust. One component
210 is an exception and is named in the data-and-code availability statement: the reproduction-check code is not in the released repository at the commit of record. The few-shot export, previously also listed as an exception, was regenerated at commit `6d7a6a71` and its validator now passes on every check. This matters given evidence that wildfire transfer conclusions are sensitive
215 to evaluation design and task formulation [45].

A companion paper treats the observational layer beneath this one, where preprocessing decisions taken before any model is fitted are shown to carry their own uncertainty budget.

Section 2 reviews the relevant literature. Section 3 describes the study
220 regions, the data, and the modelling, cross-validation, transfer and adaptation protocol. Section 4 reports results. Section 5 discusses them, including limitations. Section 6 concludes.

2. Related work

2.1. Fire susceptibility mapping, and what it reports

225 Fire susceptibility mapping has been synthesised recently from both method-
ological and application perspectives [42, 14]. The dominant pattern is sta-
ble: geospatial predictors are assembled over a study region, paired with a
historical burned-area record, and fitted with a supervised classifier, most
often a random forest [5], and the resulting surface is published with a cross-
230 validated AUC in the 0.85 to 0.95 range. That figure is a within-region
estimate. Comparable within-region results already exist for the landscapes
studied here [1, 13].

2.2. Pre-fire thermal dryness

Satellite thermal observation gives repeated access to surface state through
235 fuel moisture content [47]. Pairing land surface temperature with a vegeta-
tion index was established as a live-fuel-moisture estimator for fire-danger
rating by Chuvieco et al. [7], and the Temperature-Vegetation Dryness In-
dex [35] formalises that feature space into an internally normalised measure,
which is why it is often expected to travel between regions better than raw
240 temperature. That pre-fire thermal state carries information about subse-
quent fire is established [19, 21, 20]. Gelabert et al. [9] found dead fine fuel
moisture and its anomalies to be the most influential predictor of human-
caused ignition likelihood across Europe, testing generalisation by pooled
fitting with per-site evaluation. What has not been tested is whether a clas-
245 sifier built on this class survives strict, label-free application to a region it
never saw.

2.3. Spatial validation and transferability

Random cross-validation over spatially autocorrelated cells inflates skill estimates [33, 31], and spatially blocked designs are the standard remedy [39, 25]. The remedy is not uncontested: blocked schemes have been argued to introduce pessimistic bias of their own [43, 26, 6]. This paper uses spatial blocking throughout and reports every result at three block sizes, so that a reader can see which verdicts depend on the choice.

Blocking corrects the inflation that random folds produce within a region. It does not speak to performance on a fire the model has not seen, which Section 4.3 measures on identical cells. For that, the field’s instrument is the area of applicability and related predictor-space dissimilarity measures [23, 24, 18], which ask whether a target’s predictor values fall inside the training data’s envelope. Species distribution modelling has examined transferability far more systematically than fire has [46], and two studies there report that geographic and environmental similarity do not reliably predict transfer success [41, 34], and the transferability of satellite-derived empirical models is an active question in fire remote sensing itself [22].

2.4. Dataset shift and label-free adaptation

The canonical taxonomy separates covariate shift, where the predictor distribution moves but the predictor-response relationship holds, from concept shift, where the relationship itself changes [27]. Covariate shift is in principle correctable without target labels, by per-region standardisation or by covariance alignment such as CORAL [37], and domain adaptation has an established remote-sensing literature [38, 30]. Concept shift is not: no realignment of input distributions can repair a reversal in the sign of an

association, because detecting it requires the labels being withheld. Shift decomposition in applied remote sensing is not itself new [12]. We are not aware of a prior application of covariance alignment to fire susceptibility, fire
275 occurrence or burned-area prediction.

2.5. Cross-region generalisation of fire models

Few studies test fire-model transfer directly. Podschwit et al. [32] report, in a Peruvian case study, that meteorologically derived danger indices do not port cleanly between fire environments. WildfireGenome [16] runs
280 a leave-one-county-out matrix across seven US counties and reports strong within-county performance with highly variable off-diagonal transfer, similar pairs transferring well and dissimilar pairs collapsing; its label, however, is a principal-component composite of hazard *indicators* rather than observed burned area, and it applies no adaptation. Xu et al. [45] argue that wild-
285 fire transfer conclusions depend strongly on evaluation design, a caution we address by fixing the protocol in a project log before the diagnostics were computed and by reporting every sensitivity axis; that log is not a formal pre-registration, and Section 3.11 states what was fixed and when. Kondylatos et al. [15] provide Mesogeos, a 1 km Mediterranean datacube for data-driven
290 wildfire modelling.

Evaluation extent and AUC. Species distribution modelling settled long ago that the area a model is evaluated over is not a neutral choice. Lobo et al. [17] make it the fifth and, by their own ranking, most important reason to distrust AUC as a comparative measure: the extent of the modelled
295 area governs how many easy absences enter the calculation, and therefore the score. VanDerWal et al. [40] show the same lever on the calibration side,

that how far background points are drawn from known presences changes the fitted model and its apparent accuracy, and Barve et al. [3] give the argument its general form as the accessible area. This literature is qualitative
 300 about magnitude, because magnitude is problem-specific: it establishes that the effect exists, is directional, and can be large, without fixing a number for any given application. Two things follow for the present paper. First, our Contribution 1 is a measurement inside that framework rather than a new phenomenon, and Section 1 states it that way. Second, the wildfire literature
 305 has largely not imported the lesson: region-wide validation figures are reported as though they described performance at the fire, and we know of no wildfire study that holds the model fixed and varies only the evaluation cells. The number we obtain, 0.143 ROC-AUC, is larger than the predictor-block increments this literature reports, which is why we treat it as a reporting
 310 problem in Section 5.8 rather than a caveat.

The nearest neighbour, and the contrast the present paper draws.

Dimarco et al. [8] is the closest Mediterranean analogue: 500 m predictors harmonised across four countries, tree ensembles under spatial cross-validation, transfer tested both leave-one-country-out and as a full 4 x 4
 315 matrix, with every transfer exceeding AUC 0.80 and bioclimatically similar countries scoring higher. They apply no domain adaptation.¹

Much of the design is shared — Mediterranean regions, 500 m cells, MCD64A1-derived targets, tree ensembles, spatially aware validation, an explicit transfer matrix — and two things differ. **The predictor class:** their

¹Their Results text refers to "LST anomalies", which is inconsistent with their own

320 model rests on attributes of a place, all spatially stationary, ours on the state
of a surface in one season. **The response variable:** theirs is human-driven
ignition, ours is burned area, and those are different quantities with different
dominant controls. Read together the two results suggest that the relation-
ship between domain similarity and transfer success may be predictor-class
325 dependent, with Vesk et al. [41] and Rousseau and Betts [34] aligning from
species distribution modelling while Dimarco et al. and WildfireGenome run
against it. We present that as a live disagreement, and Section 5.7 states
why our own data do not settle it.

Methods, where no land surface temperature or TVDI variable appears
330 and the only temperature predictor is a static ERA5-Land seasonal clima-
tology [28]. We follow their Methods, which we take to be authoritative, and
note the discrepancy so that the comparison is transparent. Every statement
about their work was checked against the full text, read 2026-08-08.

3. Data and methods

335 3.1. *Study regions and temporal windows*

Five Mediterranean wildfire regions are analysed (Fig. C.1): Manavgat
2021 and Muğla 2021 in Türkiye, Bejís 2022 in Spain, North Evia 2021 in
Greece and Montiferru 2021 in Sardinia. Each is a place-based rectangular
area of interest in EPSG:4326, defined from place coverage rather than from
340 a fire perimeter, and not tuned on burned prevalence, gate outcome or any
model metric. One AOI choice was label-informed and is stated as such.
The North Evia box was extended after the legacy box was found to carry
an atypically high burned prevalence. The extended geometry was then

defined from place anchors, and the legacy variant is kept as a sensitivity
 345 arm (Appendix A). The AOIs are deliberately not clipped to fire perimeters,
 so that unburned cells around each fire form the negative class rather than
 being excluded by construction. A sixth region, Kozan 2023, is carried as a
 negative control and is excluded from all modelling by the gate of Section 3.3.

Each region has two non-overlapping windows. The **predictor window**
 350 closes the day before the **label window** opens, so no predictor observation
 can post-date the first labelled burning. Window lengths vary between re-
 gions with the event: 57 to 61 days for predictors and 35 to 59 days for labels,
 counted inclusively. A four-year baseline of window-symmetric composites
 precedes each predictor window and supplies the climatological reference for
 355 the anomaly channels. Region identifiers, bounding boxes, window dates and
 baseline years are registered in `core/regions.py` and are reproduced here.

Table 1: Study regions, areas of interest and temporal windows. Bounding boxes are in
 EPSG:4326, as registered. Window lengths in brackets are inclusive day counts. The
 baseline years are the four window-symmetric years preceding each predictor window.

Region	Bounding box (lon min, lat min, lon max, lat max)	Predictor window	Label window	Baseline years
Manavgat 2021 (Türkiye)	31.05, 36.72, 31.85, 37.35	2021-06-01 to 2021-07-27 (57 d)	2021-07-28 to 2021-08-31 (35 d)	2017, 2018, 2019, 2020
Bejís 2022 (Spain)	-1.05, 39.68, -0.35, 40.15	2022-06-15 to 2022-08-14 (61 d)	2022-08-15 to 2022-09-30 (47 d)	2018, 2019, 2020, 2021
Muğla 2021 (Türkiye)	27.1, 36.6, 28.9, 37.45	2021-06-01 to 2021-07-28 (58 d)	2021-07-29 to 2021-09-15 (49 d)	2017, 2018, 2019, 2020
North Evia 2021 (Greece)	23.05, 38.55, 23.85, 39.15	2021-06-05 to 2021-08-02 (59 d)	2021-08-03 to 2021-09-30 (59 d)	2017, 2018, 2019, 2020
Montiferru 2021 (Italy)	8.45, 40.05, 8.75, 40.27	2021-05-25 to 2021-07-23 (60 d)	2021-07-24 to 2021-08-31 (39 d)	2017, 2018, 2019, 2020

3.2. Burned-area label and the ~ 500 m analysis grid

Labels come from MODIS MCD64A1 Collection 6.1 [10] retrieved through Google Earth Engine [11], whose omission and commission characteristics [4] bound every model here. The analysis grid is **reconstructed rather than native**: the pipeline’s 30 m EPSG:4326 reference grid is partitioned into 17 x 17 blocks, giving a nominal 510 m cell that approximates rather than reproduces the MODIS cell, and is square in degrees but not on the ground. A cell’s burn date is the mode of its positive sub-pixel day-of-year values, tested against the region’s label window; the label never affects a cell’s eligibility for modelling. Two safeguards are recorded rather than assumed: cells burning inside a region’s own predictor window are removed, which ran for three of five regions, and burning in earlier years is screened for none. Appendix C.1 gives the full specification, including what follows from the grid’s shape.

3.3. Burned-landcover admissibility gate

Before any modelling each region passes a gate asking what fraction of its burned cells is dominated by natural vegetation, using ESA WorldCover classes on the same cells. A region is admitted at 0.50 with at least 30 burned cells, and rejected as a cropland-dominated control at 0.50 cropland. The purpose is to separate burned area produced by natural-fuel combustion from post-harvest stubble burning, which MCD64A1 does not distinguish. Verdicts are in Section 4.1; the full rule is in Appendix C.1.

3.4. Predictor variables

Ten predictors are used, four baseline and six thermal, all summarised per cell over the predictor window, through the chain shown in Fig. C.2.

All optical and thermal predictors come from Landsat 8 Collection 2 Level-2 (LANDSAT/LC08/C02/T1_L2), quality-screened per pixel from the QA_PIXEL band with the water bit deliberately preserved; the coarse-resolution thermal input is MODIS/061/MOD11A1.

- 385 • **NDVI**, the predictor-window median of Landsat surface reflectance, is the baseline’s one time-varying member.
- **Elevation** and **slope** come from the Copernicus DEM GLO-30.
- **Land cover** is ESA WorldCover v200 [48], entering as the dominant class code.
- 390 • **Current LST**, the predictor-window median of Landsat surface temperature in °C.
- **LST anomaly**, a z-score of the current-window LST median against the four baseline years, set to no-data where the baseline standard deviation is below 1.0 °C or observations are too few.
- 395 • **TVDI** and **TVDI difference**. The Temperature-Vegetation Dryness Index [35] is computed in the LST-NDVI feature space, with wet and dry edges taken as the 2nd and 98th LST percentiles within each of twenty NDVI bins and the index clamped to [0, 1]. The edges are percentiles of the LST values a given scene contains, so they are fitted per AOI and per window, and a TVDI of 0.5 denotes a different physical moisture state in each region. The companion paper reports what that scene dependence does and does not explain. **TVDI difference** is the raw anomaly of that index against the same four window-
- 400

symmetric baseline years used for the LST anomaly: `tvdi_difference`
 405 `= current_tvdi - mean(baseline_tvdi)`, in index units rather than
 standard deviations, which is why it is reported alongside the z-scored
 channel rather than in place of it.

- **Downscaled LST and fused LST.** A MODIS-to-Landsat downscal-
 ing model is trained on the predictor window and applied to the full 30
 410 m grid. The fused product equals observed Landsat LST wherever that
 is valid, and the downscaled surface only where it is not. Gap-filling
 therefore never replaces or blends a valid observation. The gap-filled
 share is 0.11 % to 9.70 % by region. The downscaler’s own inputs
 include coordinates, which is the one route by which a coordinate-
 415 derived surface re-enters a feature set from which Section 3.14 excludes
 coordinates. Appendix A(g) reports the increment without these two
 channels.

3.5. Cell aggregation, validity and analysis populations

Cell-level values are means over the ~ 510 m cell, computed from valid
 420 30 m pixels only, with the valid fraction recorded. A cell is `valid_for_`
`modeling` when it meets a 30 % valid-pixel floor and no thermal channel
 is missing. Two populations are then defined. The **primary** population
 is natural vegetation, cells whose combined tree, shrub and grass fraction
 reaches 0.50, which excludes cropland from every burnable mask. The **sec-**
 425 **ondary** population is all valid cells, reported throughout as a sensitivity.
 Every headline result is given for both.

3.6. Feature sets and classifier

Two nested feature sets are compared. The **baseline** set is NDVI, elevation, slope and land cover. The **thermal** set is the baseline plus the
430 six thermal channels. Land cover is one-hot encoded and numeric features are standardised inside the fitting pipeline. The classifier is a random forest [5] with 300 trees, unlimited depth, `min_samples_leaf = 3`, balanced class weights and `random_state = 42`, identical for every region, population, feature set and transfer direction, so that no comparison in this paper
435 is confounded by a model choice.

3.7. Spatial-block cross-validation and bootstrap uncertainty

Cross-validation is spatially blocked. Blocks are squares of the analysis grid formed as `row_500m // B` by `col_500m // B`, and folds are drawn over whole blocks, so no block is split between training and test. Five folds are
440 used. The block size B is reported at 2, 10 and 20 cells, about 1, 5 and 10 km, for the within-region analysis.

Uncertainty is a spatial-block bootstrap: blocks are resampled with replacement, 1000 replicates, seed 42, with 2.5 and 97.5 percentiles as the interval. Because it is blocks that are resampled, what bounds an interval's
445 reliability is the number of blocks carrying at least one burned cell, and those counts are reported alongside the intervals. Where a verdict rests on too few such blocks it is stated as indicative rather than as an interval.

3.8. Cross-region transfer protocol

For each ordered pair of regions the model is fitted on the source population and applied to the target population with **no target label used**
450

at any point: no refitting, no threshold selection, no calibration. Both the baseline and the thermal feature set are transferred, so that the thermal block’s contribution is a paired difference within a direction rather than a comparison across directions. All twenty ordered directions among the five
455 regions are computed.

3.9. Label-blind domain adaptation

Two label-free remedies are tested on every direction, in both feature sets.

Region-wise z-score. Each region’s numeric features are standardised using its own statistics, source statistics from source data and target statistics
460 from target data, never pooled. The classifier is refitted on the z-scored source and applied to the z-scored target. This removes first- and second-order marginal offsets.

CORAL after region-wise z-score. Starting from the z-scored features, the source covariance is aligned to the target covariance by the standard whitening-recolouring map [37], with regularisation $\lambda = 10^{-5}$ and eigen-
465 values floored at 10^{-12} . Critically **the transform is applied to the source only**; the target is left exactly as it is. The classifier is refitted on the aligned source. Neither variant sees a target label, and both are verified label-blind at run time.

470 λ sensitivity was assessed over nine values from 0 to 10^{-1} on four directions in both feature families, moving transfer AUC by at most 0.014 in any direction and 0.008 within the thermal family. The canonical $\lambda = 1$ lies outside the released sweep and is reported separately in Appendix A(b).

3.10. Transfer-gap decomposition and the concept-shift diagnostic

475 For each direction the gap between the target’s own within-region skill and the raw transfer result is split in two. One part is what the best label-free adaptation recovers, and the other is what it does not. The recovered fraction is defined as $(\text{adapted} - \text{raw}) / (\text{within} - \text{raw})$, signed and unclipped, and its interval comes from the same paired bootstrap. The recovered part bounds
480 what covariate-level correction can achieve. The remainder is unrecovered by that correction; Section 4.3 shows it should not be read as a conditional residual, because much of it is incurred inside a single region.

The mechanism is diagnosed by **signed univariate association**. For each numeric predictor the raw ROC-AUC of that predictor against **burned**
485 is computed in each region and never folded to $\max(\text{AUC}, 1 - \text{AUC})$, so a value below 0.5 is read as a direction rather than as weakness. A reversal is called bootstrap-supported only when the two regions’ point estimates fall on opposite sides of 0.5 **and each region’s own interval excludes 0.5**, under the same 10-cell spatial-block bootstrap. That is stricter than requiring the
490 two regions’ intervals to be disjoint: a feature whose intervals are disjoint but one of which straddles 0.5 has not been shown to point anywhere in that region, so it is recorded as a point reversal only. Appendix B states the rule again beside the counts, and `conditional_similarity_transfer.json` carries it as machine-readable metadata.

3.11. Transferability diagnostics versus transfer

495 Twenty candidate diagnostics from five families are computed for every region pair and rank-correlated (Spearman) against observed raw transfer AUC, under one bootstrap framework that resamples unordered region pairs

with both of their ordered directions travelling together. The families are
 500 marginal predictor-space measures $P(x)$, including area-of-applicability dis-
 similarity, climatic and geographic distance and a learned domain classifier;
 burned-niche overlap $P(x|y=1)$; burn-pattern regime distance $P(y)$; and con-
 ditional direction agreement $P(y|x)$. No family-wise error control is claimed,
 and the number of variants computed within each family is reported with
 505 the result. Appendix C.2 gives the full specification of each diagnostic and
 of the bootstrap.

3.12. Interventions

Pooled multi-region training. Leave-one-region-out: the model is
 trained on the pooled primary populations of the other four regions and
 510 evaluated on the held-out region, with folds blocked as before. This asks
 whether pooling recovers what single-source transfer loses.

Removal of direction-reversing features. The two predictors whose
 signed association reverses between regions with bootstrap support **on the**
frames as drawn are **elevation_mean** (Manavgat against Bejís and against
 515 Muğla) and **lst_anomaly_mean** (Bejís against Evia). Section 4.10 with-
 draws that support under an equalised frame, so this selection rule is frame-
 dependent and the arm below is reported as a measurement made under
 the original protocol rather than as a consequence of an established reversal.
 They are dropped and everything is refitted, within-region and across every
 520 direction, and the two are also dropped singly so the cost can be attributed.
 This measures what the removal costs locally and what it returns on transfer
 on the same footing. Note that the two features are selected by the same
 reversal analysis against which the result is then read, so both quantities are

post-selection estimates and no correction for that selection is applied.

525 *3.13. Controls on the transfer path*

Four evaluations establish what the transfer arms are measuring. All use the transfer protocol of Section 3.8 unchanged: fit on the source cells, apply to the target cells, no refit, no recalibration, no threshold selection. All were verified to reproduce the frozen cross-region AUCs exactly before being used.

530 **Within-region half-split.** The modelled cells of a region are cut in two at the median of a grid axis, a model is fitted on one half and applied to the other. Both axes and both directions are run, giving four splits per region. A split is discarded when either half is single-class, which happens twice in one region. Per-split source and target positive counts are reported, because
535 they are unequal and bear on the interpretation.

Foreign-region evaluation. Each held-out scar area is additionally scored with a model fitted on each of the other four regions in turn, and those four results averaged, so that a same-region model and a foreign one are compared on identical cells. This is the arm that separates the effect of
540 withholding the fire from the effect of separation distance.

Leave-one-scar-out. Burned connected components are found with 8-connectivity on the analysis grid. Every component of at least 50 cells is held out together with all cells within a buffer of it, a model is fitted on the remainder of the same region, and applied to the held-out area. Buffers of
545 2, 5 and 10 km are run. A scar held out at 2 km inside its own region is close to its training data and yet wholly unseen, which is what makes the comparison with the foreign-region arm informative. The held-out patch is defined by the labels, so this arm cannot separate the identity of a fire from

its location.

550 Appendix A(i) reports all four in full, with per-split and per-scar tables. Intervals for the four-row ladder are Student t over the eight held-out scars, which is that arm’s resampling unit, rather than the spatial-block bootstrap of Section 3.7 used everywhere else in this paper.

3.14. Leakage control

555 An explicit forbidden-column set is enforced at every model fit. Coordinates (`lon`, `lat`, `row`, `col` and their normalised forms), every burn-date and label-provenance column, and the agreement fraction are excluded from all feature sets, and the enforcement runs as an assertion rather than a convention. The natural-vegetation mask is used only to define the population, never as a predictor. Spatial blocking prevents a cell from sharing a fold with
560 its own neighbours.

Reproducibility. All randomness uses seed 42 and the bootstrap uses 1000 replicates throughout. The transfer and adaptation analysis runs in an environment separate from the upstream pipeline’s. Every within-region
565 model was therefore refitted there and compared against the frozen upstream output, and the independently implemented adaptation was compared against the pipeline’s own. The within-region comparisons agree exactly, and the twenty transfer directions agree to within 1.6×10^{-7} . Random-forest fits are not bit-identical across library versions. All numbers here were
570 therefore produced under scikit-learn 1.9.0, or verified against it. If that version is not fixed, cross-region point estimates carry an implementation tolerance of about ± 0.02 to 0.03. The companion paper reports the version sensitivity and the reproduction check in full.

Sensitivity analyses. Every headline result is repeated across two analysis populations, three spatial-block sizes, the CORAL regularisation sweep and both feature sets. Where a conclusion depends on one of those choices, the dependence is reported rather than resolved by choosing the favourable setting.

3.15. Same-geography event-to-event comparison (Muğla 2021 versus 2022)

Muğla admits a comparison in which place is held fixed and the event varies: a second fire burned inside the identical AOI, on the identical analysis grid, eleven months after the first. Signed univariate AUCs are computed for both arms under the same 10-cell spatial-block bootstrap used elsewhere. Season, year and population all differ between the arms, since the 2022 arm is defined by removing the 2021 scar, and Section 4.9 reports what that costs. Appendix C.3 gives the window dates, the population construction and the asymmetry audit.

4. Results

4.1. Study regions and the admissibility gate

All five candidate regions pass the burned-landcover gate as wildfire candidates and the negative control fails it as intended. Kozan 2023 returns a natural-vegetation fraction of 0.017 against the 0.50 threshold and is excluded from all modelling. The separation is not marginal: the five admitted regions carry fractions of 0.723 to 0.991, so the threshold falls in an empty interval rather than between neighbouring cases. That is consistent with the

gate separating burned area produced by natural-fuel combustion from post-harvest stubble burning, which MCD64A1 does not distinguish, though one negative control cannot establish it.

Table 2: Region summary: populations and gate outcomes. Counts from each region’s Step 8A dataset statistics; gate fractions from each region’s burned-landcover gate output. TSG = the primary natural-vegetation population. The TSG columns use the canonical modelled population, `burnable_tree_shrub_grass` and `valid_for_modeling == True`, which is the population every model in this paper was fitted and scored on.

Region	Total cells	Valid cells	Burned	Prevalence (all valid)	TSG cells	Burned in TSG	TSG prevalence	Burned gate verdict
Manavgat 2021	24,501	24,087	796	0.033	20,511	784	0.038	0.984 pass
Bejis 2022	15,759	15,759	1,103	0.070	15,190	1,100	0.072	0.991 pass
Mugla 2021	73,098	73,045	3,026	0.041	41,730	2,911	0.070	0.958 pass
North Evia 2021 (extended)	22,925	22,906	2,788	0.122	9,298	2,664	0.287	0.945 pass
Monftradi 2021	3,234	3,173	697	0.220	2,544	539	0.212	0.723 pass

North Evia is analysed on an extended AOI. Relative to the legacy box, the extended box leaves the burned scar essentially unchanged while cutting overall prevalence from 0.361 to 0.122 and TSG prevalence from 0.676 to 0.287. The effect of that choice on transfer is reported in Appendix A(a).

4.2. Within-region: the thermal increment replicates in five regions

Adding the six thermal predictors to the baseline raises spatially blocked
605 out-of-fold ROC-AUC in every region. The increment’s bootstrap interval
excludes zero in all five regions at 1 km and at 5 km blocking (Fig. C.3).
Those are the two scales this design supports as intervals. At 10 km the
point estimates hold, from +0.048 to +0.154. They rest on 6 to 33 positive-
carrying blocks, so they are reported as indicative rather than as intervals.
610 The table note gives the reason.

Table note (resampling units). The bootstrap resamples spatial blocks,
so what bounds an interval’s reliability is the number of blocks that carry
at least one burned cell. Those counts fall quickly as blocks coarsen. At
2 cells they are 235 (Manavgat), 302 (Bejís), 843 (Muğla), 716 (Evia) and
615 192 (Montiferru), out of 5 439, 3 967, 11 316, 2 566 and 743 blocks. At 10
cells they are 28, 19, 70, 41 and 16, out of 237, 176, 576, 155 and 35. At
20 cells they are **12, 6, 33, 15 and 6**, out of 60, 48, 167, 50 and 12. No
bootstrap replicate was invalid at any block size, with one exception. At
20 cells Bejís had 5 of 1000 replicates single-class. That is the symptom
620 of the six positive-carrying blocks just reported, and a further reason to
read the row as indicative. An equal-tailed percentile interval built on six
positive-carrying blocks has no meaningful coverage. Montiferru at 20 cells
also feeds only 12 groups into a 5-fold grouped split, so its models train on
about ten blocks each. **The 20-cell row of this table should be read**
625 **as indicative rather than as an interval.** The 10-cell row is the coarsest
blocking this design supports properly, because every region there has 16
to 70 positive-carrying blocks. The increment holds at that scale in all five

Table 3: Within-region baseline versus thermal performance and block-size robustness. Primary (TSG) population; spatially blocked 5-fold CV (Section 3.7); paired spatial-block bootstrap, 1000 replicates. Block sizes 2/10/20 cells $\approx$ 1/5/10 km.

Region	Block ($\approx$ scale)	Baseline AUC	Thermal AUC	Δ AUC	Δ AUC 95% CI
Manavgat 2021	2 ($\sim$ 1 km)	0.803	0.870	+0.067	[+0.055, +0.079]
	10 ($\sim$ 5 km)	0.748	0.797	+0.050	[+0.023, +0.077]
	20 ($\sim$ 10 km)	0.683	0.731	+0.048	[+0.014, +0.085]
Bejís 2022	2 ($\sim$ 1 km)	0.862	0.918	+0.056	[+0.048, +0.065]
	10 ($\sim$ 5 km)	0.779	0.825	+0.045	[+0.018, +0.069]
	20 ($\sim$ 10 km)	0.739	0.795	+0.057	[+0.031, +0.090]
Muğla 2021	2 ($\sim$ 1 km)	0.743	0.859	+0.116	[+0.106, +0.125]
	10 ($\sim$ 5 km)	0.698	0.777	+0.079	[+0.050, +0.105]
	20 ($\sim$ 10 km)	0.673	0.733	+0.061	[+0.030, +0.094]
North Evia 2021 (ext.)	2 ($\sim$ 1 km)	0.759	0.912	+0.153	[+0.142, +0.166]
	10 ($\sim$ 5 km)	0.716	0.864	+0.148	[+0.119, +0.183]
	20 ($\sim$ 10 km)	0.679	0.833	+0.154	[+0.124, +0.189]
Montiferu 2021	2 ($\sim$ 1 km)	0.781	0.883	+0.101	[+0.080, +0.125]
	10 ($\sim$ 5 km)	0.620	0.720	+0.100	[+0.017, +0.186]
	20 ($\sim$ 10 km)	0.555	0.681	+0.126	[+0.053, +0.228]

regions. Source: `paper/referee2_numbers.md`, block C, counted from the frozen per-cell prediction tables.

630 This within-region result is not itself novel. It is reported because the transfer arms below are measured against it.

4.3. What the evaluation frame is worth, within one region

This section isolates the effect that Section 4.10 later applies between regions. It is reported first because it is measured on one model in one
635 region with no transfer involved, so it cannot be explained by anything the transfer arms do.

Where the skill is lost, on a matched comparison. Four evaluations are reported. The last three are scored on **identical cells**, so they differ only in what the model was trained on, and the first is included to show why an
640 unmatched comparison misleads. Each uses the transfer protocol unchanged: fit, then apply with no refit, no recalibration and no threshold selection.

The held-out unit is a burned connected component of at least 50 cells together with all cells within 2 km of it. What makes that a harder discrimination problem than a whole region is the composition of its negatives, not
645 its burned fraction, and that can be shown rather than argued. Taking the same fitted model and the same out-of-fold predictions, and scoring them on a random sample of region cells drawn at the scar area’s own burned fraction, gives **0.782 against the region-wide 0.782**: matching the prevalence changes nothing, at $-0.000 [-0.003, +0.002]$ over the nine scars. Scoring the
650 same predictions on the scar area itself gives 0.627, a fall of **+0.155 [+0.093, +0.217]**. The whole effect is therefore the negative pool. Every negative in a scar collar is fire-adjacent, sharing the terrain, land cover and synoptic

conditions of the positives, whereas a region's negatives include its easy far field. The burned fraction, 34 to 87 % against 3.8 to 28.7 % for a region, is
655 a symptom of that construction, and ROC-AUC is in any case invariant to class balance at fixed class-conditional distributions.

Table 4: The four evaluations, scored on identical cells. Primary natural-vegetation population. Rows B, C and D are scored on the held-out scar area; row A is the whole region and is shown to make the mismatch visible. Means and Student t intervals are over the eight held-out scars.

Evaluation	Model trained on	Scored on	Mean AUC	95 % CI
A. Blocked cross-validation, 5 km	the region, scar included	the whole region	0.776	[0.738, 0.814]
B. Same blocked model, restricted	the region, scar included	the scar area	0.634	[0.552, 0.716]
C. Leave-one- scar-out	the region, scar withheld	the scar area	0.552	[0.501, 0.602]
D. Foreign region	another region, 306 to 2,802 km	the scar area	0.555	[0.495, 0.616]

All four rows are means over the **same eight scars**. A ninth burned component, Bejís, is excluded throughout: it is that region's only component of any size, so holding it out leaves no usable source model and there is no
660 row C for it. Rows A, B and D are reported here on the eight so that the differences are paired, and the intervals are Student t over those eight, which is the resampling unit for this arm rather than the spatial-block bootstrap used elsewhere in the paper. Eight is a small number and the intervals are wide accordingly. Four of the eight scars are in Muğla and two in Montiferru,
665 so they are not independent; row A in particular is a region-level quantity

repeated across the scars of a region, and its interval is pseudo-replicated and should not be read as coverage. The two differences are computed per scar and paired, which is what they are reported for.

The same model, scored two ways on the same region, differs by
670 **0.143 AUC.** Rows A and B are one model. The only change is which cells it is scored on: the whole region, or the burn scar and its 2 km collar. That change alone costs **0.143** [**+0.077**, **+0.208**] of the 0.225 [**+0.157**, **+0.293**] fall from A to C, about two thirds, and it is the size of the entire increment this literature usually reports. Both intervals are Student *t* over the eight
675 scars, so the "two thirds" is a ratio of two estimates and is compatible with anything from roughly a third to nine tenths; the point of the comparison is the size of the numerator, not the precision of the fraction. A region-wide blocked figure is therefore an upper bound on what the same model achieves where the fire actually is, and the difference is not small enough to ignore.
680 Any comparison of a scar-level result against a region-level reference inherits that, and the region-level reference is what a paper of this kind normally reports.

Neither withholding the fire nor moving 2,800 km has a measurable cost. B minus C, the fire-specific residual on identical cells, is **+0.082**
685 [**-0.011**, **+0.175**], which spans zero. C minus D, the effect of replacing a same-region model with one fitted between 306 and 2,802 km away, is **-0.003** [**-0.075**, **+0.069**], which also spans zero and is an order of magnitude smaller. The second is the cleaner result: on the same cells, a model fitted in another country does as well as one fitted in the same region with
690 the scar withheld, and it survives restriction to Muğla, the only region where

holding out one scar still leaves the source model properly trained, at 0.579 against 0.597.

Row D is a mean over four foreign models, and the spread behind it is large. Decomposing it gives **32** scar-by-source combinations, eight scars by four foreign sources, running from 0.374 to 0.724, with **ten of the 32 below chance** and a mean spread of 0.171 across the four sources for a given scar, reaching 0.329 for one. The pooled 0.555 is therefore not a statement that any foreign model does as well as a same-region one. It is a statement that the *average* foreign model does, and the variation it averages over is the pair-specific sign instability that Sections 4.5 to 4.9 are about. Both facts belong together: a region boundary costs nothing on average, and which side of that average a particular pair lands on is not predictable from anything measured here.

The first difference should not be over-read, and one confound has to be stated. Holding out a region's only large scar also removes most of its positives: the source model retains 11 positives in Evia, 88 in Manavgat and 97 in Montiferru, against about 2,000 in Muğla, which has four separate scars. Those four non-Muğla arms average 0.525 and the four Muğla arms 0.579. Three of the four are genuinely starved; the fourth, Montiferru's second component, retains 472 source positives, so the shortfall there is not a training-size effect. so row C mixes "the fire was withheld" with "almost all the positives were withheld". Restricted to Muğla, C is 0.579 and D is 0.597, and the C to D comparison still shows nothing. With eight scars, three of them starved, this design cannot establish a fire-specific residual, only bound it at about 0.18 against the 0.143 that separates row A from row B on the

same model.

What this establishes is bounded rather than positive. Rows C and D both sit close to chance, so a comparison between them has little dynamic range, and neither the fire’s identity nor the region boundary is shown to
720 cost anything on this evaluation. What is measurable is the change of evaluation geometry between rows A and B. What it does not establish is the fire event as the unit, because the held-out patch is defined by the labels, so its identity cannot be separated from its location. Two further observations point the same way: a model trained on the other half of the same fire is no
725 better at matched sample size, and the within-region half-split, which does see half of the target scar, returns 0.574 against leave-one-scar-out’s 0.552 with overlapping intervals.

Separation does not order the twenty-direction matrix either. The Spearman correlation between centroid separation and transfer AUC is -0.32 with
730 an interval spanning zero, and the two nearest directions, Manavgat and Muğla at 306 km, are among the worst at 0.470 and 0.401. This is a different quantity from Table 9’s geographic-distance row, computed on the twelve-direction subset, which gives -0.24 ; both span zero. The transfer mean of 0.541 is over all twenty directions and is not the value at any one separation:
735 the 2,802 km pair returns 0.326 and 0.444.

The result is robust to how the held-out patch is defined, at 0.543 to 0.565 across minimum component sizes from 25 to 200 cells and under both 4- and 8-connectivity, and flat across buffers of 2, 5 and 10 km.

The increment declines with the holdout, and is not established
740 **once the fire is withheld.** Contribution 1 is about the paired thermal-

minus-baseline difference, so the same evaluations were run on it.

Table 5:

Evaluation	Thermal increment	Interval
Blocked cross-validation, per region	+0.056 to +0.153	every interval above zero
Within-region half-split	+0.027	positive in 13 of 18 splits
Leave-one-scar-out	+0.022	[−0.032, +0.077]
Cross-region, twenty directions	+0.004	[−0.028, +0.036]

The point estimate falls monotonically as the holdout hardens, and the last two intervals span zero. Under leave-one-scar-out the increment is positive in six of eight scars and averages +0.022, against a baseline of 0.529 and a thermal 0.552 that are both close to chance on that evaluation; restricted to Muğla it is +0.009 [−0.051, +0.070]. **The within-region increment of +0.056 to +0.153 is therefore substantially a property of interleaved holdout.** It is not shown to be zero on an unseen fire, and eight scars cannot show that; what this design establishes is that it is not established there.

4.4. Cross-region transfer, and what label-free adaptation does to it

Raw transfer is heterogeneous and includes anti-predictive directions. Raw target AUC spans 0.326 to 0.686 (Fig. C.4). Twelve of 20 directions are above chance with CI support. Six are *below* chance with CI support: both directions of Manavgat to Bejís and of Manavgat to Muğla, plus Bejís to Evia and Evia to Bejís. Two intervals span 0.5. Even the best

raw transfer, Evia to Manavgat at 0.686, stays far below that target’s own within-region thermal performance of 0.870. Across all directions the raw deficit against the within-region reference is 0.184 to 0.592 AUC. That reference is a region-level blocked estimate; the controls below show it is not
760 matched to a transfer evaluation.

Those counts belong to the 2-cell blocking of Table 7. At the more conservative 10-cell blocking the same points give 9 above, 4 below and 7 uncertain, and no direction changes side of the chance line (Section 4.8). Four of the
765 six below-chance directions keep their support there. Bejís to Manavgat and Manavgat to Muğla lose it and carry no verdict. The qualitative statement is unchanged. The counts should not be read as exact.

In precision terms the transfer is worse than the ROC figures suggest. ROC-AUC is the metric used throughout this paper, for comparability with the literature, but a susceptibility surface is used as a ranked
770 area budget, so precision-recall is the operational quantity. Across the twenty directions the thermal model’s PR-AUC averages **0.156 against a no-skill baseline of 0.136**. The mean of the twenty per-direction lifts is 1.16; the ratio of the two means just quoted is 1.146. **Six of the twenty fall below their own no-skill baseline at the point estimate, and five of those six have intervals entirely below it.** The exception is Bejís to Manavgat, at 0.034 [0.029, 0.042] against a baseline of 0.038, whose interval covers the baseline and which is therefore a point-estimate case only. Only one direction, Evia to Manavgat, exceeds twice its baseline, at 0.094 against 0.038.
775
780 A transferred model therefore ranks burned cells about a sixth better than random on average, and worse than random in five directions with interval

support and a sixth at the point. This is a sharper statement than the ROC means support and it should be the one a practitioner reads.

The static baseline does not transfer either. This is the control
785 for the reading the rest of the paper invites, and it constrains that reading sharply. The same twenty directions were run with the terrain, fuel and greenness baseline alone. The mean target AUC is **0.537**, against **0.541** for the thermal model. Static attributes of a place are the class Dimarco et al. transfer successfully, and are the class this paper’s framing treats as
790 portable. Here that class is itself barely above chance. The paired per-direction contrast is given in Section 4.7b. What matters here is that the transfer failure below is not specific to the dynamic block. A baseline that does not travel, plus pre-fire thermal state, gives a model that does not travel.

The thermal block’s paired contribution to transfer, with its in-
795 **terval.** The two matrices were differenced direction by direction. The mean is **+0.004**, but the mean is not the informative statistic here. The individual paired contributions span **-0.148 to +0.133**, with **twelve positive and eight negative** (`baseline_vs_thermal_transfer.csv`). The spread is thirty times the mean, and the sign is not a property of the block but of
800 the pair it is asked to cross: the same six predictors that add +0.133 in one direction subtract 0.148 in another. A mean near zero here records cancellation, not consistent absence of effect. The directions are not independent, because each region appears in eight of the twenty. The interval therefore depends on what is treated as the resampling unit. All four units the design
805 permits give the same answer:

Every interval spans zero, and the point estimate is a small fraction of

Table 6:

Resampling unit	n	95 % interval on the mean paired contribution
Directions, naive	20	$[-0.027, +0.034]$
Unordered pairs, cluster bootstrap	10	$[-0.028, +0.036]$
Unordered pairs, t on pair means	10	$[-0.034, +0.042]$
Regions, leave-one-out jackknife	5	$[-0.037, +0.046]$

each width. The per-region jackknife shows how little the mean is anchored. Holding out Manavgat, Bejís, Muğla, Evia and Montiferru in turn gives +0.0148, +0.0021, +0.0065, **-0.0081** and +0.0060. **Dropping Evia alone**
810 **reverses the sign of the headline.** None of these four units propagates within-direction sampling variability. They resample between directions only.

On the frames as drawn, six directions are below chance with interval support, the sharpest at 0.326 [0.305, 0.349]. No account of merely lost skill produces a reliably reversed ranking, so on this evidence those six
815 would need a mechanism acting on the direction of the relationship, and Sections 4.5 to 4.9 pursue one. **Section 4.10 then shows that most of this count is an artefact of the evaluation frames: equalising them leaves one direction below chance, not six.** The reader should carry that forward through the intervening sections, whose reversal evidence
820 is computed on the frames as drawn. Per-split and per-scar detail is in Appendix A(i).

Label-blind adaptation compresses the matrix toward chance rather than repairing it. Under region-wise z-scoring the twenty directions span 0.431 to 0.630 and under CORAL 0.443 to 0.624, roughly half

Table 7: Cross-region transfer matrix, thermal model, TSG population. Target ROC-AUC with 2-cell spatial-block bootstrap 95% CIs (1000 replicates). CORAL is applied after region-wise z-scoring ($\lambda = 10^{-5}$).

Direction	Raw	Region-wise z-score	CORAL
Manavgat→Bejís	0.326 [0.305, 0.349]	0.477 [0.451, 0.502]	0.511 [0.484, 0.534]
Bejís→Manavgat	0.444 [0.408, 0.480]	0.457 [0.420, 0.497]	0.555 [0.528, 0.583]
Manavgat→Muğla	0.470 [0.451, 0.488]	0.431 [0.411, 0.449]	0.443 [0.423, 0.461]
Muğla→Manavgat	0.401 [0.378, 0.426]	0.559 [0.531, 0.587]	0.560 [0.535, 0.587]
Manavgat→Evia	0.613 [0.593, 0.631]	0.542 [0.520, 0.565]	0.539 [0.518, 0.561]
Evia→Manavgat	0.686 [0.653, 0.716]	0.516 [0.489, 0.544]	0.527 [0.500, 0.553]
Bejís→Muğla	0.618 [0.601, 0.635]	0.518 [0.501, 0.535]	0.507 [0.489, 0.524]
Muğla→Bejís	0.583 [0.561, 0.607]	0.535 [0.512, 0.557]	0.560 [0.538, 0.581]
Bejís→Evia	0.383 [0.363, 0.402]	0.532 [0.509, 0.551]	0.499 [0.479, 0.518]
Evia→Bejís	0.448 [0.426, 0.470]	0.549 [0.524, 0.575]	0.549 [0.525, 0.573]
Muğla→Evia	0.653 [0.636, 0.671]	0.561 [0.543, 0.580]	0.563 [0.545, 0.582]
Evia→Muğla	0.577 [0.560, 0.593]	0.501 [0.485, 0.518]	0.530 [0.515, 0.546]
Montiferru→Manavgat	0.567 [0.539, 0.594]	0.573 [0.538, 0.609]	0.606 [0.574, 0.639]
Manavgat→Montiferru	0.533 [0.488, 0.580]	0.586 [0.540, 0.629]	0.592 [0.550, 0.631]
Montiferru→Bejís	0.548 [0.521, 0.578]	0.574 [0.552, 0.596]	0.569 [0.548, 0.591]
Bejís→Montiferru	0.594 [0.560, 0.631]	0.550 [0.500, 0.601]	0.574 [0.530, 0.621]
Montiferru→Muğla	0.619 [0.604, 0.634]	0.576 [0.562, 0.589]	0.565 [0.550, 0.579]
Muğla→Montiferru	0.531 [0.495, 0.568]	0.587 [0.549, 0.624]	0.584 [0.547, 0.623]
Montiferru→Evia	0.586 [0.565, 0.606]	0.630 [0.611, 0.649]	0.624 [0.605, 0.641]
Evia→Montiferru	0.647 [0.608, 0.682]	0.568 [0.528, 0.609]	0.581 [0.539, 0.623]

825 the raw spread, with no adapted direction exceeding 0.631 against the unmatched within-region references of 0.859 to 0.918. Adaptation raises the failing directions and degrades most of those that already transferred. Taking the better of the two adaptations per direction, 14 of the 20 end closer to chance than they began and 6 end further from it; five of those six involve
830 Montiferru, the smallest and last-added region, and move upward, while the sixth is Manavgat→Muğla moving downward from 0.470 to 0.443. The 14 to 6 split should be read at the precision of limitation (ix) in Section 5.9, since Bejís→Manavgat is counted as compressed on a margin of 0.001.

Against the right reference, adaptation is not failing (Fig. C.5).

835 The three controls above give an achievable reference for a model applied to a fire it has not seen: 0.574 for the half-split, 0.552 for the leave-one-scar-out. CORAL, committed to in advance, averages **0.552** across the twenty directions against 0.541 raw, which is exactly that reference. Taking whichever of the two adaptations scores higher per direction gives 0.556, but that selection uses the target labels the protocol forbids, so it is an oracle upper
840 bound rather than an achievable result and is reported as one. What it does is regress the matrix towards that value: fourteen of twenty directions move closer to chance, which harms the directions that already worked and helps the ones that did not. The verdict that follows is narrower than "alignment fails". Alignment cannot exceed what a model can achieve on an unseen fire,
845 and it does not.

In the six directions where raw transfer was below chance, the best label-free method recovers at most 34 % of the gap to the within-region reference, so the remaining unrecovered fraction is at least 0.66 everywhere. The half-

Table 8: Transfer-gap decomposition (four-AOI set, 12 directions). Within = target’s within-region thermal AUC; best adapted = the better of z-score/CORAL; recovered fraction = (adapted – raw)/(within – raw), signed and unclipped, with paired bootstrap CI (1000 replicates). Montiferru directions are not part of this decomposition (per-pair absolute decompositions exist without fraction CIs). The status column asks whether the *adapted* value clears chance and uses the 2-cell adapted intervals of Table 7. The adapted arms were not recomputed at the coarser blocking of Appendix A(c), which covers the raw arm and the paired delta only.

Direction	Within	Raw	Best adapted (method)	Recovered fraction [CI]	Status
Manavgat→Bejís	0.918	0.326	0.511 (CORAL)	+0.31 [+0.28, +0.34]	recovery, chance not excluded
Bejís→Manavgat	0.870	0.444	0.555 (CORAL)	+0.26 [+0.18, +0.34]	recovery above chance
Muğla→Manavgat	0.870	0.401	0.560 (CORAL)	+0.34 [+0.28, +0.40]	recovery above chance
Bejís→Evia	0.912	0.383	0.532 (z-score)	+0.28 [+0.23, +0.32]	recovery above chance
Evia→Bejís	0.918	0.448	0.549 (z-score)	+0.22 [+0.16, +0.27]	recovery above chance
Manavgat→Muğla	0.859	0.470	0.443 (CORAL)	−0.07 [−0.12, −0.03]	negative recovery
Muğla→Bejís	0.918	0.583	0.560 (CORAL)	−0.07 [−0.16, +0.01]	negative recovery
Manavgat→Evia	0.912	0.613	0.542 (z-score)	−0.23 [−0.32, −0.14]	negative recovery
Evia→Manavgat	0.870	0.686	0.527 (CORAL)	−0.86 [−1.18, −0.60]	negative recovery
Evia→Muğla	0.859	0.577	0.530 (CORAL)	−0.17 [−0.22, −0.11]	negative recovery
Muğla→Evia	0.912	0.653	0.563 (CORAL)	−0.35 [−0.43, −0.27]	negative recovery
Bejís→Muğla	0.859	0.618	0.518 (z-score)	−0.42 [−0.51, −0.34]	negative recovery

850 split control above shows that the larger part of that remainder is already
incurred inside the region, so it should not be read as a measure of concept
shift. Seven directions show *negative* recovery, meaning adaptation moves
the score away from the reference; in six of those raw transfer was already
above chance and adaptation destroyed that advantage. Label-free alignment
855 therefore does not act as a repair mechanism.

4.5. Transferability diagnostics: what appears to order transfer, and why it does not

Twenty candidate diagnostics from five families were each rank-correlated
with the same target quantity, the raw thermal transfer AUC over the twenty
860 ordered directions, under one common pair-based bootstrap.

**Only two diagnostics have intervals excluding zero, and both
are conditional.** The stronger is the sign-agreement fraction over interval-
supported features, at $\rho = +0.84$ [$+0.58, +0.88$]. The cosine variant reaches
 $+0.81$. No marginal measure was shown to order the matrix. That includes
865 area-of-applicability-style dissimilarity in predictor space, climatic distance
and geographic distance. The niche-overlap and regime families were not
shown to order it either. The learned domain classifier is at ceiling, separating
source from target at $\text{AUC} \geq 0.96$ for every pair. It always succeeds, which
is why it carries no ordering information.

870 Two limits are stated with the result rather than after it. The first is
size. The index's tie structure caps the achievable Spearman at $+0.861$,
so the observed $+0.840$ sits essentially on that ceiling. Its exact one-sided
permutation p is 0.0060 , the smallest this tie structure can produce, against
a Bonferroni threshold of 0.0026 over the nineteen computed variants. No

Table 9: Transferability diagnostics versus raw thermal transfer, by family. Spearman ρ against raw transfer AUC with pair-based bootstrap 95 % CIs. Exp. is the sign expected if the diagnostic orders transfer. The member named is the one with the largest absolute correlation in its family, which is not always in the expected direction. All twenty individual diagnostics are in Appendix B, Table B1.

Family	Diagnostics	Exp.	Largest correlation in family	Sign as expected	Any CI excluding 0
P(y x) condi- tional	6	+	+0.84 [+0.58 , +0.88] agree- ment fraction, supported features	yes	yes, 2 of 6
P(x y=1) niche overlap	5	+	+0.24 [−0.45, +0.74] Schoener’s D, 1-D mean	yes	no
P(x) marginal	6	−	−0.32 [−0.78, +0.33] domain- classifier AUC	yes	no
P(y) regime structure	2	−	+0.29 [−0.38, +0.74] ^{log} ₄₄ effective- N distance	no	no
geographic	1	−	−0.24 [−0.84,	yes	no

875 outcome of this diagnostic could have cleared family-wise correction on ten
effective pairs. The second limit is labels. Signed associations need burned
labels in both regions, so the family that appears to work is not available
before deployment, while the family that fails is.

A fourth limit removes the result entirely, and is reported in
880 **Section 4.10.** The sign-agreement index is built from the signed associations
that Section 4.10 shows to be artefacts of the evaluation frames, and it was
correlated against transfer measured on those same frames. Recomputed on
an equalised frame it is unanimous, taking the value 1.0 in every direction
with no variance left to correlate, while the continuous cosine variant falls
885 from $\rho = +0.50$ to $+0.12$. Everything in this section is therefore reported as
what the original protocol yields, and the conclusion that survives is that **no**
diagnostic tested here was shown to order transfer once the frames
are comparable.

A third limit is about selection, and it is the sharpest of the three. The
890 two rows that clear zero are the *supported-feature* variants. Their feature
subset is chosen by whether two regions' bootstrap intervals happen to be
disjoint. That is a data-dependent selection, made on the same data, with
no correction. The unselected counterparts over all nine features are $\rho =$
 $+0.50$ $[-0.17, +0.83]$ for the cosine and $\rho = +0.18$ $[-0.40, +0.72]$ for the
895 agreement count. Both span zero. The result lives in the selection step, and
is reported as such.

The families are compared on unequal samples, and equalis-
ing them does not change the ordering. The marginal, applicability,
climatic and geographic rows sit on twelve directions, because those diag-

900 nostics exist only for the four-region subset. The supported-conditional rows sit on sixteen and the rest on twenty. A reader may reasonably ask whether the marginal family’s failure is a statement about power rather than about diagnostics. Every row was therefore recomputed on the common twelve directions. The published values reproduce to 4.9×10^{-5} . The conditional rows still lead, at $+0.87 [+0.65, +0.88]$ and $+0.85 [+0.43, +0.88]$, and every 905 marginal row still spans zero. The ordering is not an artefact of unequal samples.

4.6. The contrast pair: similarity is not sufficient

The clearest single view of Table 9 needs no ranking at all (Fig. C.8). 910 Manavgat and Muğla lie in the same country and the same fire year. They are 306 km apart by the centroid geodesic distance this paper uses as a diagnostic, and their nearest boundaries are 191 km apart. Their burned cells occupy the most similar environmental envelope of any pair in the matrix, with per-feature Schoener’s D of 0.77 to 0.89. Yet on the frames as drawn 915 five of nine feature-response directions point opposite ways. Elevation is one of them, at a signed AUC of 0.374 [0.289, 0.471] in Manavgat against 0.611 [0.532, 0.690] in Muğla, with disjoint intervals, and all four absolute thermal channels are among the others. **Section 4.10 withdraws that reading:** this is the pair whose Manavgat arm is 60 % far field, and under a 10 km collar 920 the elevation figure moves to 0.561 against 0.606, on the same side of 0.5. The contrast that survives is the transfer result itself, not the reversal count that was offered to explain it. Transfer is below chance in both directions at the point estimate, at 0.470 and 0.401. Both directions also sit inside the nominal area of applicability, at 0.875 and 0.531 of target cells.

925 Bejís and Montiferru sit at the opposite extreme. Their burned envelopes barely overlap, and they carry the most dissimilar values on every overlap measure. They transfer above chance in both directions. That half of the contrast rests on point estimates: neither direction carries a verdict at 5 km blocking, and the marginal applicability audit was never produced for 930 Montiferru. The claim is one of *sufficiency*. Similarity does not guarantee transfer, and dissimilarity does not preclude it. The claim rests on two coexisting counterexamples, so it does not depend on the number of pairs available.

4.7. Interventions: pooling and feature removal

935 **(a) Pooled multi-region training** (Fig. C.6). At the point estimate, training on the pooled primary populations of the other four regions never beats the best single-source transfer for any target, with shortfalls of 0.02 to 0.22, and it stays 0.28 to 0.50 AUC below the within-region ceiling; for two targets it falls below the pairwise mean and below chance, though only Bejís 940 is below chance with interval support, at 0.417 [0.369, 0.467]. Aggregation does not recover what single-source transfer loses.

(b) Removing the direction-reversing features. The two predictors whose signed association reverses between regions **with bootstrap support on the frames as drawn** are **elevation_mean** and **lst_anomaly_mean**. 945 Section 4.10 withdraws that support under an equalised frame, so this selection rule is frame-dependent and the arm below measures what the removal costs under the original protocol, not a consequence of an established reversal. The per-region signed associations and the three pair-level reversals that meet that criterion are in Appendix B, Tables B2 and B3. Retraining

950 without them costs **-0.081** of mean within-region AUC, supported in every region (per-region deltas -0.060 , -0.130 , -0.074 , -0.063 , -0.079 , every interval entirely below zero), and changes mean transfer by **+0.014** [**-0.017**, **+0.045**], which spans zero.

Two qualifications belong with those numbers rather than after them. First, the debit is not the thermal block's. Dropping elevation alone accounts 955 for -0.061 of it, from 0.888 to 0.827 . Dropping the LST anomaly alone accounts for -0.013 , from 0.888 to 0.875 . Roughly three quarters of the cost is therefore the removal of a **baseline** terrain variable, not of a thermal channel. Second, both figures are post-selection. The two predictors were 960 chosen because they reverse, using the same data on which the -0.081 and the $+0.014$ are then estimated. No correction for that selection is applied.

Table 10:

Configuration	Mean within-region AUC	Mean transfer AUC
full	0.888	0.541
drop <code>elevation_mean</code>	0.827	0.546
drop <code>lst_anomaly_mean</code>	0.875	0.544
drop both	0.807	0.556

Fig. C.7 shows the four configurations together. What this measures is a local cost with no compensating transfer gain. It does not measure an exchange, because the transfer side is a null on both arms. The thermal 965 block's own contribution to transfer is $+0.004$, with an interval spanning zero (Section 4.4). Removing the reversing predictors returns $+0.014$, with an interval spanning zero. Two nulls on the portability axis are not a price paid.

4.8. Sensitivity analyses

970 Eight design choices were varied with everything else held fixed. They are the Evia AOI and its prevalence, the CORAL regularisation constant, the blocking scale, the closure date of the predictor window, the quality screening of the coarse thermal input, the contrast between the normalised and the absolute dryness channels, the removal of the coordinate-informed
975 channels, and the capacity of the classifier. None changes a conclusion above. Two bound how the results should be read, so they are carried into the main text here.

Coarsening the blocks from 1 km to 5 km moves the **paired thermal-minus-baseline delta** verdicts from ten positive, seven negative and three
980 uncertain to six, four and ten. The above-chance verdicts on the thermal arm itself, a different quantity, move from twelve, six and two to nine, four and seven. Support is removed from seven verdicts and added to none. The point estimates are unchanged, but that is an identity rather than a result. The blocking scale is the resampling unit, and it cannot move an estimate
985 computed once over all target cells.

Manavgat’s whole downstream chain was then rebuilt from a quality-screened MODIS input. That changes the downscaled surface on 22,304 of 24,150 cells, by up to 10.9 °C. No signed univariate association moves by more than +0.0003. This closes the one processing-artefact candidate for
990 that region’s behaviour. Appendix A reports all eight arms, including one that tests a claim of Section 1.2 and does not uphold it.

4.9. *The same geography, a second fire: reversal with place held constant*

Every result above compares different places. Muğla admits a stricter test, because a second fire burned inside the identical AOI on the identical analysis
995 grid. The 2021 event has a predictor window closing on 28 July. The 2022 event has a matched window closing on 20 June. Region, bounding box, cell definition, feature registry and processing chain are the same in both arms. The design holds place fixed. It does not hold season fixed, because the 2022 fire ignites about six weeks earlier, and it does not hold the population fixed,
1000 because the 2022 arm is defined by removing the 2021 scar.

Elevation reverses with bootstrap support. In 2021 higher ground burned preferentially, at 0.611 with its interval entirely above 0.5; in 2022 lower ground did, at 0.296 with its interval entirely below. The two intervals are disjoint, 0.532 against 0.355, and the difference is -0.317 $[-0.414,$
1005 $-0.220]$. The same predictor, the same region, the same grid, and an association that points the opposite way in two fires eleven months apart. The four absolute thermal channels change side as well, from bootstrap-supported lower-values-burn in 2021 to higher-values-burn point estimates in 2022.

Two structural properties must be read alongside these numbers, because
1010 they have no analogue in the twenty-direction matrix. The two arms are not disjoint samples: they share 38,789 of the 2022 arm’s 38,790 cells, and elevation, slope and land cover are identical to the digit across all 73,098 grid cells, so only NDVI and the six thermal channels carry new information between them. And the removal is the target’s own positive class, so in the
1015 2022 to 2021 direction not one target positive is present in the source training population while 38,789 of 38,819 target negatives are, and membership of

Table 11: Signed univariate feature-burned AUC, Muğla 2021 versus 2022. Raw AUC against **burned**, never folded to $\max(\text{AUC}, 1 - \text{AUC})$; 10-cell (≈ 5 km) spatial-block bootstrap, 1,000 replicates, seed 42 (Section 3.15). Analysis population 41,730 rows / 2,911 burned (2021) and 38,790 rows / 331 burned (2022). **Positive-carrying 5 km blocks: 70 for the 2021 arm and 11 for the 2022 arm.** Table 3’s note sets sixteen as the floor this design supports at that blocking, so the 2022 intervals here fall below the paper’s own standard and are read as indicative, exactly as the 20-cell row of Table 3 is. The 2022 arm is additionally a single compact scar, so its eleven blocks are contiguous. Read from `paper/step9g_raw/.../mugla_2021__mugla_2022_event_relative/step9g_direction_reversal_table.csv`.

Feature	2021 AUC [95 % CI]	2022 AUC [95 % CI]	Reversal
elevation_mean	0.611 [0.532, 0.690]	0.296 [0.230, 0.355]	bootstrap- supported
current_lst_mean	0.325 [0.271, 0.382]	0.515 [0.434, 0.580]	point only
current_tvdi_mean	0.336 [0.275, 0.398]	0.594 [0.475, 0.674]	point only
downscaled_lst_mean	0.307 [0.253, 0.366]	0.508 [0.435, 0.571]	point only
fused_lst_mean	0.325 [0.272, 0.383]	0.519 [0.436, 0.583]	point only
ndvi_mean	0.662 [0.616, 0.704]	0.707 [0.624, 0.777]	none
slope_mean	0.637 [0.582, 0.686]	0.558 [0.468, 0.634]	none
lst_anomaly_mean	0.485 [0.395, 0.566]	0.380 [0.249, 0.502]	none
tvdi_difference_mean	0.490 [0.396, 0.575]	0.397 [0.265, 0.514]	none

the source training set alone separates the 2021 target’s classes at ROC-AUC 0.9996. We therefore cannot bound what that asymmetry does to a transfer estimate between the two arms, which is why none is reported here, only state that a direction with this structure is not comparable to the twenty between-region directions. The elevation reversal is the finding; the structural asymmetry is a competing explanation this design cannot exclude. The known part of the bias runs the safe way: the removed cells are high, with a median of 563 m, and unburned in 2022, so removing them raises the 2022 elevation AUC and makes the reversal smaller rather than larger.

4.10. The evaluation frame, applied to our own matrix

Section 4.3 established that swapping a region’s negatives for fire-adjacent ones moves ROC-AUC by +0.155 [+0.093, +0.217] with the model held fixed. That result was obtained inside one region. It applies with equal force between regions, and applying it changes what Sections 4.3 and 4.5 can claim. This section reports that test. Source: `aoi_frame_auc.csv`, `aoi_frame_transfer.csv`, `paper/code/verify_aoi_frame.py`, `paper/code/verify_aoi_transfer.py`.

The five areas of interest are not comparable frames. Each region is a rectangle drawn around a fire, and the rectangles differ by an order of magnitude in how much unburnt far field they enclose.

Montiferru’s frame is fire-scale; Manavgat’s and Bejís’s are roughly three-fifths far field. The far field is not a neutral addition. In Manavgat the median elevation of modelled cells rises from 472 m within 5 km of the fire to 955 m at 10 to 20 km and 1,273 m at 20 to 50 km, against 512 m for the burned cells themselves, in Taurus terrain that no plausible spread model would place at

Table 12: Evaluation-frame geometry of the five study regions. Primary natural-vegetation population. Distance is Euclidean to the nearest burned cell on the 500 m grid, at 0.45 km per cell. Source `aoi_frame_auc.csv`; recomputable by `paper/code/verify_aoi_frame.py`.

Region	cells	burned	median distance to burned	share beyond 10 km
Manavgat	20,511	784	13.4 km	60.1 %
Bejís	15,190	1,100	13.5 km	63.1 %
Muğla	41,730	2,911	11.3 km	55.3 %
Evia	9,298	2,664	8.0 km	43.7 %
Montiferru	2,544	539	2.7 km	2.1 %

risk.

Under an equalised frame the sign reversals of Section 4.6 do **not survive**. Restricting every region to cells within 10 km of any burned
1045 cell drops **no positives**, only far-field negatives.

On the equalised frame all five regions agree in sign on elevation, on LST and on TVDI, and both bootstrap-supported elevation reversals of Table B3 disappear. Applying this paper’s own criterion from Section 3.10, which requires each region’s own 10-cell block-bootstrap interval to exclude 0.5, **no**
1050 **reversal remains bootstrap-supported in the collar frame** (`collar_frame_bootstrap.csv`). Two features straddle 0.5 at the point estimate and neither is supported. `lst_anomaly_mean` runs 0.392 [0.324, 0.460] in Bejís against 0.584 [**0.497**, 0.669] in Evia, so the point estimates fall on opposite sides but Evia’s interval includes 0.5 by 0.003. `tvdi_difference_`
1055 **mean** straddles as well, at 0.509 [0.417, 0.597] in Muğla against 0.384 [0.288,

Table 13: Signed univariate AUC, frame as drawn against a 10 km collar. Point estimates; the intervals that decide the reversal question are given in the text below and in `collar_frame_bootstrap.csv` (10-cell blocks, 1000 replicates, seed 42). Signed and never folded to $\max(\text{AUC}, 1 - \text{AUC})$, so a value below 0.5 is a direction, not weakness. The collar drops no burned cells in any region.

Signed univariate AUC	Manavgat	Bejís	Muğla	Evia	Montiferru	straddles 0.5
elevation, full frame (Table B2)	0.374	0.643	0.611	0.541	0.584	yes
elevation, 10 km collar	0.561	0.614	0.606	0.648	0.581	no
current LST, full frame	0.538	0.477	0.325	0.377	0.370	yes
current LST, 10 km collar	0.386	0.405	0.332	0.286	0.376	no
current TVDI, full frame	0.552	0.517	0.336	0.362	0.356	yes
current TVDI, 10 km collar	0.392	0.454	0.342	0.250	0.361	no

0.504] in Montiferru, and no region’s interval excludes 0.5. These are the two internally differenced channels, and an earlier draft of this section treated the first as uniquely informative because it carries no lapse-rate signal; that argument does not hold, because the second is equally decorrelated from elevation and behaves the same way. Both are point reversals, not supported ones. We state it that way rather than quoting the point estimates alone, because the alternative would be to apply a looser standard to the arm that supersedes Table B3 than to Table B3 itself. Elevation under the collar is above 0.5 in all five regions but individually supported in only two, Muğla at 0.606 [0.525, 0.685] and Evia at 0.648 [0.550, 0.740].

The sign the five regions agree on is not the one the dryness framing predicts. For LST the common direction is *below* 0.5 in every region, at 0.386, 0.405, 0.332, 0.286 and 0.376: a hotter pre-fire surface is associated with **less** burning, not more, and the same holds for TVDI. This is not a lapse-rate artefact. Pooling concordance within elevation deciles leaves LST at 0.402, 0.509, 0.363, 0.327 and 0.392, still below 0.5 in four of five regions, and within NDVI deciles at 0.440, 0.486, 0.404, 0.279 and 0.368, below 0.5 in all five, despite $r(\text{LST}, \text{NDVI})$ reaching -0.92 . The likely reading is that fuel availability dominates surface dryness at this scale in this cohort: NDVI itself is the most consistently supported single predictor under the collar, at 0.582 to 0.663 with intervals excluding 0.5 in four of five regions, more than any other feature, though in Muğla and Evia the absolute thermal channels are further from 0.5, so greener and therefore more fuel-rich cells burn more, while the hottest cells are the sparse, rocky ones with little to burn. Section 1.2 motivates the thermal block from moisture-stress physics;

that motivation is not what the signed associations show, and Section 5.2 states the consequence.

The same test destroys the one diagnostic that appeared to work. Section 4.5’s only diagnostic with an interval excluding zero is the
 1085 fraction of features whose signed association points the same way in source and target, counted over features interval-supported in both. That statistic is built out of exactly the signed AUCs this section has just shown to be frame artefacts, and it was correlated against transfer measured on the same unequal frames. Recomputing both sides under the 10 km collar (`diagnostics_`
 1090 `collar_frame.csv`, `paper/code/verify_diag_collar.py`):

Table 14: The two diagnostics that ordered transfer, recomputed on an equalised frame. Spearman ρ against target ROC-AUC over the ordered directions in which the diagnostic is defined. Source `diagnostics_collar_frame.csv`; recomputable by `paper/code/verify_diag_collar.py`.

Diagnostic	Full frame	10 km collar
Sign-agreement fraction over supported features	$\rho = +0.86$ ($p = 0.0001$, $n = 14$), values 0, 0.5, 1	1.0 in all 18 defined directions, variance exactly 0 — degenerate
Cosine similarity over all nine signed AUCs	$\rho = +0.50$ ($p = 0.023$, $n = 20$)	$\rho = +0.12$ ($p = 0.61$, $n = 20$)

Once the frames are equalised every region pair agrees in sign on every jointly supported feature, so the diagnostic has no variance left and its correlation with transfer is undefined rather than weak. This is not a marginal shift: the number of features supported in both regions *rises* from 1.20 to

1095 3.40 per direction, so the diagnostic is better determined and unanimous.
The disagreements it was reading were the far fields.

Contribution 3 is therefore restated, and it becomes a stronger negative result. The finding is not that conditional similarity orders transfer where marginal similarity fails. It is that **no diagnostic tested here,**
1100 **marginal or conditional, orders transfer once the evaluation frames are equalised** — and the one that appeared to was measuring how the study rectangles were drawn. The practitioner’s position is worse than Section 4.5 implied, not better: there is no screen, and the apparent exception was an artefact. Section 4.5’s numbers are retained as computed, on the frames as
1105 drawn, because they are what a reader following the original protocol would obtain; this section is the correction.

Three further points bound how many independent reversals could have been counted. Within the collar, `fused_1st_mean` correlates with `current_1st_mean` at 0.99 to 1.00 and `downscaled_1st_mean` at 0.97 to 0.99, and
1110 `current_tvdi_mean` at 0.87 to 0.98. The four absolute channels are close to one axis, and the two anomaly channels correlate at 0.64 to 0.94, so the "five of nine directions reverse" count of Section 4.6 is a count of features, not of independent quantities; in effective dimensions it is closer to two. The channels whose reversal vanishes are the ones that were partly measuring terrain. Current LST correlates with elevation at $r = -0.695, -0.125,$
1115 $-0.404, -0.511$ and -0.507 across the five regions, and TVDI at -0.722 to -0.298 , so their proxy strength varies fivefold between regions and a frame that shifts the elevation distribution shifts them with it. The two internally differenced channels are decorrelated from elevation by construction — the

1120 LST anomaly at -0.195 to $+0.256$ — and neither of them yields a supported reversal either, so this explains which reversals were manufactured without licensing a claim that any survived.

The transfer matrix moves as well. Restricting source and target to the same collar:

Table 15: Cross-region transfer under equalised evaluation frames. Primary natural-vegetation population, thermal model, twenty ordered directions per row. Above/below chance are point counts; the supported counts use the same 10-cell (≈ 5 km) spatial-block bootstrap on the target as Table 7, 1000 replicates, seed 42. Per-direction bounds are in `aoi_frame_transfer.csv`.

Source frame	Target frame	Mean target AUC	Above chance	Below chance	Supported above / be- low	Paired thermal delta
full	full (Table 7)	0.540	14 of 20	6	9 / 4	+0.003
full	10 km	0.575	17 of 20	3	11 / 1	+0.002
10 km	full	0.571	17 of 20	3	11 / 3	+0.014
10 km	10 km	0.617	19 of 20	1	15 / 1	+0.023
5 km	5 km	0.608	18 of 20	2	12 / 0	+0.014

1125 The reference arm reproduces the frozen matrix, at 0.540 against Table 7’s 0.541 and 14 of 20 exactly, so this is measuring the same quantity. The above-

and below-chance counts in this table are point counts. Under the same 10-cell block bootstrap used for Table 7, at 1000 replicates, the full frame gives nine directions above chance and four below with interval support, and the collar frame fifteen above and one below, so the headline movement is six to one at the point estimate and **four to one with interval support** (`aoi_frame_transfer.csv`, which carries the per-direction bounds). The largest movers are Bejís to Evia, 0.383 to 0.602, and Bejís to Manavgat, 0.440 to 0.601.

What this does and does not change. Three claims do not survive and are withdrawn here rather than defended. The count of six anti-predictive directions becomes one. The sign reversal of elevation, LST and TVDI as the mechanism of the residual is a property of the frames, not of the predictor-burning relationship. And the paired thermal contribution is not stable at +0.004: it rises to +0.023 once frames are equalised, so the claim that the block contributes nothing between regions is frame-dependent too. What survives is the central negative result, and it survives with room to spare: equalised transfer of 0.617 against within-region skill of about 0.87 leaves most of the gap intact. Section 5.9 records the frame as a limitation of the cohort rather than of the method, since equalising it is cheap and we recommend it.

This is the paper’s own Contribution 1 turned on the paper’s own matrix. We had measured that the composition of the negative pool moves ROC-AUC by more than the effects this literature reports, and had then computed a five-region transfer matrix, and a mechanism, on frames whose fire-adjacent share ranges from 37 % to 98 %. We report it because it is the strongest

available demonstration that the reporting standard of Section 5.8 is not a technicality.

4.11. What target labels cost: the recovery curve

1155 Everything above measures a failure; this prices it. Label-free alignment does not close the residual gap, so the missing resource is information about the target that alignment cannot synthesise, and the direct way to supply it is target labels. A frozen labelled-budget diagnostic answers how many, for three regions across all six ordered directions, using one 10-cell (~ 5 km) 1160 spatial block as the unit of labelling effort and reading recovery against a matched target-only ceiling of 0.777 to 0.824.

Thirty-two labelled blocks recover **85 to 89 % of that ceiling in three of the six directions**, two of which started below chance; 51 to 57 % in two more, the directions where the conditional gap is widest; and 30 % in the 1165 sixth. That budget is 7 to 20 % of the target's natural-vegetation population, which at the 2,700 to 3,000 labelled cells those blocks carry (S1.3) and this grid's effective cell area of 0.199 to 0.208 km² (Appendix C) is roughly **540 to 620 km²**, so it is a real answer and not a cheap one. Two properties of the measurement bound it and are stated in the supplement rather than 1170 buried: at the top budget the selection pool is nearly exhausted, so the narrow upper-budget intervals reflect saturation rather than precision, and the labelled blocks are drawn from the event being predicted, which is not a resource available before that event burns. The protocol, the full curve and the limits are in Supplement S1.

1175 Two limits belong with the number. At small budgets the same intervention damages the direction that already transfers best without any labels.

And six directions in three regions cannot support a general label budget, so none is offered.

5. Discussion

1180 5.1. *Principal findings*

Three results carry this paper, each established in Section 4 and not re-tabulated here.

First, where a model is scored decides what it appears to know. Changing only the evaluation cells costs 0.143 ROC-AUC, the cause is the composition
1185 of the negative pool rather than prevalence, and applying the same correction between regions withdraws three claims we had made from the frames as drawn, including the sign reversal we had offered as the mechanism of the transfer residual (Section 4.10).

Second, the thermal block’s local skill does not travel, and the frame
1190 correction does not rescue it: equalised transfer sits at 0.617 against within-region skill near 0.87, and the static baseline travels no better than the dynamic one, so this is not a peculiarity of thermal predictors.

Third, no diagnostic was shown to order the transfer matrix. Two appeared to, both requiring burned labels in the target region and both resting
1195 on a data-selected subset of predictors, and the frame correction of the first finding removes even those: recomputed on comparable frames the winning index is unanimous and has no variance left to correlate.

5.2. *Why the thermal increment is real but local*

A distinction has to be drawn before this section can say anything useful.
1200 The within-region increment and the transfer failure are measured at different

separations, and Section 4.3 shows that most of the difference between them is already present inside a single region. What follows therefore reads the increment as local in a specific sense: it holds where the held-out cells are interleaved with training cells, and most of it is gone once they are not, before the fire or the region changes. What that does not explain is why the directions differ from each other, and in particular why six of them are anti-predictive.

The within-region increment is not an artefact to be explained away, but it is measured under a holdout that Section 4.3 shows is generous. Withholding a whole burn scar reduces it to $+0.022$ $[-0.032, +0.077]$, so what follows describes an increment that is established under interleaved validation and not established beyond it. It replicates in five independent regions, survives coarsening of the spatial blocks to ~ 5 km with its interval intact, persists in the secondary all-valid population, and survives a predictor window closed 7 and 14 days earlier in all five regions. The direction of that last result is region-specific and should not be overstated: the contribution strengthens in Bejís ($0.058 \rightarrow 0.079$ at 14 days) and Muğla ($0.115 \rightarrow 0.128$), is flat in Montiferru, and **weakens monotonically in Evia** ($0.156 \rightarrow 0.149 \rightarrow 0.135$). What holds everywhere is survival, not improvement. The transfer failure is therefore not evidence that the thermal signal is spurious. It is evidence that the fitted relationship is *local*.

The natural objection is that different Mediterranean regions are simply different systems, so that a predictor meaning one thing in one place and another elsewhere is not instability but a comparison of two different systems. The two Muğla events answer it directly: region, AOI, grid, feature registry

and processing chain identical, static predictors identical cell by cell, and elevation’s association with burning reversing from 0.611 [0.532, 0.690] to 0.296 [0.230, 0.355], a difference of -0.317 [-0.414 , -0.220] (Section 4.9). Holding geography fixed does not stabilise the direction of the relationship.

1230 Three caveats attach and none is dismissed. Season and year are confounded, and **that confound cannot be resolved in this study area**: the events sit 42 days apart in median burn day-of-year, neither year contains a second event at the other’s phase, and a calendar-matched arm would carry nine burned cells against a gate minimum of thirty. The population is
1235 not held fixed either, the 2022 arm being the 2021 arm with the 2021 scar removed. And the 2022 arm rests on one compact scar and eleven positive-carrying 5 km blocks. What survives all three is the narrower reading that *something* reverses the elevation-burning association with geography, grid and pipeline held fixed, which is why Section 4.9 is reported as corroboration
1240 rather than as the load-bearing evidence for the mechanism.

On the frames as drawn the sharpest supported reversal belongs to elevation, a static predictor, so instability is a property of the predictor-to-burning mapping generally rather than of thermal channels specifically. This is the same conclusion the baseline transfer arm reaches from the other direction:
1245 the instability is not the thermal block’s peculiarity.

5.3. *Why the similarity-based diagnostics fail*

A diagnostic built on distance in predictor space asks one question: do the target’s predictor values look like the training data’s? That is not the question that matters when the failure is conditional. A target region can
1250 sit well inside the training envelope while the relationship between those

predictors and burning points the other way. Manavgat to Muğla is exactly that case, with 0.875 of target cells inside the weighted area of applicability and transfer below chance.

The same holds for the niche-overlap and regime families. At the point
1255 estimates, the pair with the highest burned-niche overlap in the matrix
fails in both directions while the pair with the lowest transfers in both, so
no monotone function of overlap can order the outcomes. The qualifier is
not decorative: at the 5 km blocking this paper otherwise defends, neither
Bejís↔Montiferru direction carries a verdict and Manavgat→Muğla loses its
1260 below-chance support, so this contrast is a statement about point estimates
and is made as one. The domain classifier is at ceiling for every pair, which
is informative in itself: the regions are trivially separable in predictor space
whether or not their transfer works, so separability carries no ordering infor-
mation here.

1265 5.4. *What the conditional diagnostic is, and what it is not*

The sign-agreement index reaches $\rho = +0.84$ [+0.58, +0.88] against ob-
served transfer. Three limits are reported with it rather than after it, and
the sharpest is a selection: the two rows that clear zero are supported-feature
variants whose subset is chosen on the same data, and their all-nine-feature
1270 counterparts span zero. Its tie structure caps the achievable Spearman at
+0.861, so the observed value sits essentially on its own ceiling and the rank-
ing it supports is coarse. Its exact one-sided permutation p is 0.0060, the
smallest that tie structure can produce, against a Bonferroni threshold of
0.0026 over nineteen computed variants. No outcome could have cleared
1275 family-wise correction on ten effective pairs. And it needs burned labels

in both regions, which makes it a mechanism diagnosis rather than a pre-deployment screen.

A fourth limit removes it altogether. The index is built from signed associations that Section 4.10 shows to be artefacts of the evaluation frames, and recomputing it on an equalised frame leaves it unanimous and variance-free, with the continuous variant falling from $\rho = +0.50$ to $+0.12$. So the paper's practical conclusion is not "use this index instead", and it is not even "the index works but needs labels". It is that **none of the twenty candidates was shown to order transfer once the frames are comparable**, and the one that appeared to was reading how the study rectangles were drawn. That is a worse position for a practitioner than Section 4.5 first suggests, and it is the honest one. As always these are nulls on ten effective pairs: not shown to order transfer, rather than shown incapable of it.

5.5. What the two interventions do and do not show

Both interventions show the same shape. Pooling four regions never beats the best single-source transfer for any target and stays 0.28 to 0.50 below the within-region ceiling, so aggregation does not manufacture the missing conditional information. Removing the reversing predictors costs -0.081 of within-region skill with interval support in every region, and changes mean transfer by $+0.014$ $[-0.017, +0.045]$, which is not distinguishable from zero.

That pair of numbers is easy to read as an exchange, and it is not one. The thermal block's own contribution to transfer is $+0.004$, with an interval spanning zero. Removing the reversing predictors returns $+0.014$, with an interval spanning zero. Both arms are null on the portability axis. What the interventions measure is a local cost and no compensating transfer gain.

That is not a conservation law, and it is not a rate at which local skill can be sold for portability. No such rate is estimated here.

5.6. *The regime hypothesis, reported as it happened*

A regime-structure explanation was stated in advance and the data confirmed the null. The regime-distance correlation has the wrong sign at the point estimate, the most regime-similar pair fails in both directions, and the most regime-different pair transfers above chance. One mundane explanation can be set aside: cutting Muğla, by far the largest population, to Manavgat's cell count and separately to its positive count leaves Muğla's transfer behaviour inside the subsampling range in both roles. The error was in the hypothesised grouping, not in the data, and with ten pairs this cannot refute regime typology [2] in general.

5.7. *The empirical contrast with Dimarco et al.*

Dimarco et al. [8] transfer successfully across a comparable Mediterranean design and we do not, and the two results are not in conflict. Their predictors are attributes of a place, ours describe the state of a surface in one season, which suggests that the relationship between domain similarity and transfer success is predictor-class dependent.

That reading is one of at least two, and we cannot separate them here. Their response variable is human-driven **ignition**; ours is burned **area**. Ignition likelihood is dominated by human access and activity, stable and near-universal drivers, whereas burned extent is dominated by spread. A predictor-class explanation and a response-variable explanation are therefore confounded in this comparison. Our own data speak against a simple

1325 predictor-class reading in any case: the static baseline transfers at a mean
of 0.537 here, so within this cohort the place-attribute class does not travel
either.

5.8. *Implications*

For practice, one number should be stated before any other. In precision-
1330 recall terms, which is how a susceptibility surface is actually used, transferred
models average a PR-AUC of 0.156 against a no-skill baseline of 0.136, and
six of twenty directions fall below their own baseline. Whatever the ROC
figures suggest, a model moved to a region it was not fitted in does not
usefully rank burned cells there.

1335 This paper also supports one concrete change in what is reported. Along-
side a spatially blocked within-region figure, report skill on a held-out burn
scar and its surroundings, or on a held-out fire event. On these five regions
the two differ by about 0.14 ROC-AUC on the same model, which is the size
of the effect such papers usually claim, so a blocked figure alone should be
1340 read as an upper bound. Transfer skill likewise has to be *measured* rather
than inferred from similarity, and the pre-deployment diagnostics currently
used for that inference did not order it here. Where a model must be moved,
the resource that closes the gap is target labels: thirty-two labelled 5 km
blocks recover 85 to 89 % of the target's matched ceiling in three of six direc-
1345 tions, and 30 to 57 % in the rest. That is a real answer and not a cheap one,
being 7 to 20 % of the target's natural-vegetation population, and at small
budgets the same intervention damages the direction that already transfers
best.

For method development, the results bound what unsupervised alignment

1350 can be asked to do. Two label-free methods applied carefully moved fourteen
of twenty directions towards chance, and five of the six they moved upward
involve Montiferru, the smallest region; the sixth, Manavgat to Muğla, moves
downward. The committed-in-advance CORAL arm reaches a mean of 0.552,
and taking whichever of the two methods scores better per direction reaches
1355 0.556 — but that selection uses the target labels the protocol forbids, so it is
an oracle upper bound rather than an achievable result (Section 4.3). Even
the oracle only reaches the reference a model can reach on an unseen scar at
all (Section 4.3), so they are not failing far below an achievable target; they
are regressing the matrix onto it, which costs the directions that already
1360 worked. What alignment cannot do is exceed that reference, and a sign
reversal is not a distribution mismatch that realigning inputs would repair.

5.9. Limitations

The transfer failure is a finding, not a limitation. The limitations are the
boundaries on how far it generalises.

1365 (i) **No meteorological covariates** enter the models, so we cannot say
how local skill and portability behave for a mixed thermal-plus-weather pre-
dictor set.

(ii) **The same-geography comparison covers one region only**, and
even there year and seasonal phase are confounded, a confound that cannot
1370 be resolved in this study area for the reason given in Section 5.2. Its 331
burned cells also leave the thermal reversals unresolved at interval level,
and the pair holds place fixed but not population. Those two arms are
additionally the only transfer directions here computed by us rather than
read from the pipeline author’s frozen export, with his unmodified code and

1375 the same pinned environment.

(iii) **All labels derive from a single burned-area product**, MCD64A1 [10], whose omission and commission characteristics [4] bound every model evaluated here. No second burned-area product covers 2021 and 2022 at this resolution; the companion paper reports what an independent active-fire observation says about the omission concern.

(iv) **Evia remains the most imbalance-atypical population** even after the AOI extension, at a TSG prevalence of 0.287 against 0.038 to 0.072 in three of the others.

(v) **Each region contributes one fire season**, so regional concept shift is confounded with event meteorology, and distinguishing them requires multi-year labels.

(vi) **Cross-region point estimates carry an implementation tolerance** of roughly ± 0.02 to 0.03 across scikit-learn versions. All reported numbers are fixed to one verified version, but exact reproduction elsewhere requires the archived environment.

(vii) **The diagnostic correlations rest on an effective sample of ten region pairs**. Both the successes and the failures of Section 4.5 should be read at that power.

(viii) **Manavgat’s atypical transfer behaviour remains unexplained**. It is where the conditional diagnosis bites hardest and where feature removal recovers most. Three candidates have now been tested and none survives: its meteorology, which was not extreme; the quality screening of its coarse thermal input, which propagates widely but moves no signed association by more than +0.0003 (Section 4.8, Appendix A(e)); and the evaluation frame, which

1400 explains its elevation figure but not its transfer behaviour (Section 4.10).
With one fire season per region the remaining candidates are not separable
in this design.

(ix) **The interval-support counts are less stable than the point estimates behind them.** Several verdicts sit within a thousandth of their
1405 reference value, and at 1 km blocking the published split of ten positive,
seven negative and three uncertain turns on a lower bound of -0.00045 . The
point estimates and the sign pattern are stable; the counts are not. Every
sentence in this paper that leans on an exact count of supported directions
should be read at that precision. (x) **The five areas of interest are**
1410 **not comparable frames, and this cohort cannot fully repair it.** Each
region is a rectangle drawn around a fire, and the share of modelled cells lying
beyond 10 km of any burned cell ranges from 2 % in Montiferru to 63 % in
Bejís. Section 4.10 shows this is not cosmetic: equalising the frame to a 10
km collar removes both supported elevation reversals, makes all five regions
1415 agree in sign on elevation, LST and TVDI, lifts mean transfer from 0.540 to
0.617 and reduces the below-chance directions from six to one. We report the
equalised arm alongside the frame-as-drawn arm rather than replacing one
with the other, because the collar radius is itself a choice and 5 km and 10 km
do not agree exactly (0.608 against 0.617). The deeper limitation is that the
1420 frames were fixed upstream of this work, in **repo/**, so we can restrict them but
not extend them; a region whose rectangle is already fire-scale, Montiferru,
cannot be given a far field for symmetry. Any future cohort should fix the
frame by an explicit accessible-area rule [3] before any predictor is computed,
and we treat that as the main design lesson of this paper.

1425 (xi) **One classifier family.** The headline numbers use a random forest with unlimited depth, the configuration most able to encode local structure and least able to extrapolate. Appendix A(h) shows the transfer result is not an artefact of that choice: three further estimators, including a penalised linear one, all land between 0.510 and 0.556 and all place fourteen of twenty
1430 directions above chance. Those are point estimates without intervals, so the ordering among them is not claimed as a result. Other model families were not tried, and a different inductive bias might behave differently, but within this family the negative result is a property of the predictors rather than of an unregularised estimator.

1435 6. Conclusions

Where a fire model is scored decides what it appears to know. Holding the model, its predictors and its fitting fixed and changing only which cells are scored, moving from a whole study region to the burn scar and its 2 km collar costs **0.143 ROC-AUC** [**+0.077**, **+0.208**]. A control isolates the
1440 cause: matching prevalence changes nothing, at -0.000 [-0.003 , $+0.002$], while scoring the same predictions on the scar area costs $+0.155$ [$+0.093$, $+0.217$]. It is the composition of the negative pool, not class balance. The effect is larger than the predictor-block increments this literature publishes as findings, so a region-wide validation figure should be read as an upper
1445 bound on what a model achieves where fire actually occurs, and reported as one.

That correction is not only other people's problem. Applied to our own five-region matrix it withdraws three of our claims. The study areas enclose

very different far fields, from 2 % to 63 % of cells beyond 10 km of any fire,
1450 and that far field is higher ground. Equalising them to a 10 km collar, which
drops no burned cells, lifts mean transfer from 0.540 to 0.617, reduces the
directions below chance from six to one, and makes all five regions agree in
sign on elevation, LST and TVDI. The sign reversal we had offered as the
mechanism of the transfer residual is therefore mostly a property of how five
1455 rectangles were drawn. Under our own criterion no reversal then remains
bootstrap-supported. The sign the five regions agree on is also not the one
the dryness framing predicts: hotter pre-fire surfaces burned less everywhere,
surviving stratification within elevation and greenness, so on this cohort the
thermal block behaves as a proxy for fuel availability.

1460 What survives the correction is the central result. Pre-fire thermal dry-
ness adds a real and repeatable within-region increment, +0.06 to +0.15
ROC-AUC over a static and near-static baseline in all five regions, surviv-
ing spatial blocking at about 5 km and a predictor window closed up to
two weeks before the first labelled burning. It does not travel. Equalised
1465 transfer of 0.617 against within-region skill near 0.87 leaves most of the gap
intact; the paired cross-region contribution is +0.004 $[-0.028, +0.036]$ with
a sign that varies by pair; and the static baseline transfers at 0.537 against
the thermal model's 0.541, so this is not a peculiarity of dynamic predictors.
Removing the two reversing predictors costs -0.081 of within-region skill,
1470 three quarters of it elevation, and changes transfer by +0.014, whose interval
spans zero: a local cost is measured, no compensating transfer gain is, and
no exchange between them is demonstrated.

The shortfall cannot be anticipated. None of the twenty diagnostics tested

ordered the matrix on the marginal, niche or regime families, and on ten
1475 effective pairs those nulls mean not shown to order transfer rather than shown
not to. Two conditional variants did carry intervals excluding zero, but they
rest on a data-selected subset of predictors, require burned labels on both
sides, and — on the equalised frames of the first finding — become unanimous
and variance-free, so they were reading the study rectangles rather than
1480 the transfer. No diagnostic tested here survives as a screen. Label-free
alignment compresses fourteen of twenty directions towards chance rather
than repairing them, and pooled multi-region training does not escape it
either. Supervised recalibration works but is not cheap: thirty-two labelled
5 km blocks recover 85 to 89 % of the target’s matched ceiling in three of six
1485 directions, and at the top budget the labelled set already holds most of one
region’s burned cells.

Two practical consequences follow. Report local skill and portability sep-
arately, since a predictor block worth a large within-region increment may
contribute nothing distinguishable from zero across regions. And fix the
1490 evaluation frame by an explicit accessible-area rule before any predictor is
computed, because a frame chosen for convenience can manufacture both
a transfer failure and a mechanism to explain it. Three extensions are left
for future work: temporal transfer, meteorological covariates, and physically
normalised dryness variables. Each is a step towards the self-calibrating
1495 thermal monitoring system that motivated this study.

Appendix A. Appendix A. Sensitivity analyses

Each arm below varies one design choice and leaves everything else fixed. None changes a conclusion in the main text. They are reported so that a reader can see which choices were tested, and what each was worth. Arm (f) is the exception in one respect: it is not a robustness check but a test of a claim the introduction makes, and the claim is not upheld.

(a) Evia AOI and prevalence. The raw transfer arms were repeated with the legacy, high-prevalence Evia box. Every qualitative conclusion is unchanged. Thermal raw transfer AUCs move by up to 0.07, the largest being Evia to Bejís at 0.378 to 0.448, and no direction changes side of the chance line.

(b) CORAL regularisation, including the canonical $\lambda = 1$. Over nine λ values from 0 to 10^{-1} on four directions, CORAL transfer AUC moves by at most 0.014 within any direction and 0.008 within the thermal family. The canonical $\lambda = 1$ of the cited method lies outside that sweep and was computed separately on the same four directions. It changes the thermal mean from 0.519 to 0.522, a shift of +0.003 with a per-direction range of -0.005 to $+0.016$, and none of those four directions crosses the chance line. The $\lambda = 1$ values are 0.510, 0.444, 0.559 and 0.575 against 0.508, 0.445, 0.564 and 0.559 at $\lambda = 0.1$.

Those four directions are Bejís–Muğla and Manavgat–Muğla, and they exclude the pair that moves. An earlier version of this appendix generalised from them to the whole matrix, which was an error: the Manavgat–Bejís pair is not in the sweep and it is the pair whose CORAL result is λ -dependent. A separate frozen artefact, `experiments/cross_`

region/step10/coral_lambda_sensitivity.csv, carries it. Bejís to Manavgat gives 0.5571 [0.5292, 0.5857] at $\lambda = 10^{-5}$, 0.5644 at $\lambda = 10^{-3}$ and 0.5473 at $\lambda = 0.1$, all with intervals entirely above 0.5, but **0.4850 [0.4496, 0.5197] at $\lambda = 1$** , which crosses the chance line and loses its interval support.

1525 Manavgat to Bejís falls likewise, from 0.5108 to 0.4776 [0.4501, 0.5013].

The correct statement is therefore narrower. CORAL-dependent conclusions are insensitive to λ over three orders of magnitude below 0.1; at the cited method's canonical $\lambda = 1$ the one direction whose adapted interval sits entirely above chance loses that status. $\lambda = 10^{-5}$ is the minimally regularised
1530 choice and matches the reference implementation, but the dependence is real and is reported rather than absorbed.

(c) **Blocking scale.** Recomputing the transfer quantities at 10-cell ($\approx$ 5 km) blocking from the frozen per-cell predictions widens the intervals and moves the verdict counts, from ten positive, seven negative and three uncertain at 1 km to six, four and ten at 5 km. Coarser blocking therefore removes
1535 support from seven verdicts and adds none. Across five seeds the counts run 5 to 6 positive, 3 to 4 negative and 10 to 11 uncertain, so they are not exact.

The point estimates are unchanged. That is an identity rather than a result, and it should not be offered as robustness. The blocking scale is
1540 the bootstrap *resampling unit*, and each point estimate is computed once over all target cells, so no choice of block size could have moved one. The comparison does establish two things. Every verdict that changes, changes towards "no verdict". And no direction crosses the chance line under the widened intervals. The counts are the fragile part of this paper. The sign
1545 pattern is not thereby shown to be robust. It is simply not tested by this

variation.

(d) Predictor-window closure. The predictor window was closed 7 and 14 days earlier in all five regions. The thermal contribution stays positive and bootstrap-supported everywhere. The direction of the change is region-specific. It strengthens in Bejís, from 0.058 to 0.079 at 14 days, and in Muğla, from 0.115 to 0.128. It is flat in Montiferru. It weakens monotonically in Evia, from 0.156 to 0.149 to 0.135. What holds in every region is survival, not improvement. That is the claim made in Section 5.2.

(e) Quality screening of the coarse thermal input. Two of the five regions' MODIS inputs are quality-screened and three are not. The split follows export date rather than design, and it induces an elevation-correlated change at the input, at $r = +0.615$ in Manavgat. Manavgat's entire downstream chain was rebuilt from a quality-screened input. That changes the downscaled surface on 22,304 of 24,150 cells, by up to 10.9 °C. No signed univariate association moves by more than +0.0003. Elevation is identical in both arms, at 0.374, the population is unchanged, and the within-region increment moves from $[+0.055, +0.079]$ to $[+0.054, +0.077]$. The reason is structural. Elevation is a DEM variable the screening cannot touch, and fusion falls back on the MODIS-derived surface across only 2.14 percentage points of coverage. Details are in `paper/modis_qc_downstream_propagation.md`.

(f) Normalised against absolute dryness channels. Section 1.2 argues that an internally normalised index should be less exposed to absolute-temperature offsets between regions than raw land surface temperature. The thermal block contains both kinds, so the argument can be tested directly as

a feature-set contrast.

One thing should be said before the result, because it makes the outcome less surprising than it might otherwise appear. The normalised channels are the two that hold their direction in the same-geography two-event comparison of Section 4.9, where the absolute channels move. That is a statement
1575 about one region across two fires. **Across regions it does not hold:** `1st_anomaly_mean` is itself one of the two predictors whose reversal is bootstrap-supported on the frames as drawn, reversing between Bejís and Evia (Section 4.10 withdraws that support under an equalised frame), which is why
1580 Section 4.7b drops it alongside elevation. Being internally normalised protects a channel against the offset between two seasons in one place. It does not, on this evidence, protect it against a change of place.

Three feature sets were run over all twenty directions and all five within-region folds, with the classifier, population, folds and bootstrap held fixed.
1585 The harness aborts unless its reference configuration lands on the frozen exports. It did exactly: the maximum absolute difference between the reference arm and the frozen transfer AUCs is 0.000000 across all twenty directions.

The advantage is not found. Per direction, the normalised set minus the absolute set has a mean of -0.0037 . It is positive in 12 of 20 directions and
1590 runs from -0.099 to $+0.094$, so the spread is an order of magnitude larger than the difference. The normalised channels transfer very slightly worse at the mean, and nothing here supports a design rule favouring anomaly-referenced dryness for portability.

Within region the two sub-blocks are indistinguishable, at 0.8551 and
1595 0.8564, and each recovers most of the gap between the baseline’s 0.7896 and

Table A.16:

Feature set	Mean transfer AUC	Directions > 0.5	Mean within-region AUC
baseline only	0.5371	16	0.7896
baseline + normalised anomalies	0.5442	15	0.8551
baseline + absolute surface state	0.5479	13	0.8564
all ten features (reference)	0.5414	14	0.8883

the full model’s 0.8883. What separates them in sign stability does not become transferable skill.

This arm strengthens the negative finding rather than softening it. Four differently constituted feature sets were tried. None clears a mean of 0.548 across regions, and all differ sharply within them. The baseline arm also gives an independent confirmation of the control reported in Section 4.4: its mean transfer of 0.5371 was recomputed here from the modelling datasets and matches the 0.537 read from the frozen per-direction export. Details are in `paper/anomaly_only_transfer.md`.

The comparison is between two sub-blocks of one thermal set on one cohort. It does not test normalised dryness indices in general, and it does not test a normalisation fitted against a pooled multi-region reference rather

than each region’s own baseline years.

(h) Model capacity. Every number in this paper comes from one ran-
1610 dom forest with unlimited depth and `min_samples_leaf = 3`. That is the
configuration most able to encode local structure and least able to extrapo-
late, so a reader may reasonably ask whether the transfer failure is a property
of the predictors or of the estimator. Holding the model fixed is right for the
internal comparisons and does not license the external claim, so three fur-
1615 ther estimators were run over the same twenty directions and the same five
within-region folds.

Regularisation does not rescue transfer. All four estimators land
between 0.510 and 0.556, all four put exactly fourteen of twenty directions
above chance, and the linear model, the one built to extrapolate, transfers
1620 worst. Within-region skill falls as capacity is reduced, from 0.888 to 0.741,
which is what regularisation is expected to cost; transfer does not rise to
meet it. The thermal block’s contribution to transfer is +0.004 under the
canonical forest and **negative** under all three regularised alternatives, so the
block does not become portable when the model is made simpler.

1625 The transfer failure is therefore a property of the predictors on this cohort,
not of an unregularised forest. Detail in `paper/model_capacity.json`.

(g) The coordinate-informed channels. `downscaled_1st_mean` and
`fused_1st_mean` come from a per-region downscaling model whose own in-
puts include coordinates, which Section 3.14 names as the one route by which
1630 a coordinate-derived surface re-enters a feature set that excludes coordinates.
A coordinate-smoothed surface is by construction locally informative and
non-portable, so if the within-region increment depended on it, that incre-

Table A.17:

Estimator	Within-region AUC	Within increment	Transfer AUC	Transfer increment	Above chance
Random for- est, depth un- lim- ited, leaf 3	0.888	+0.099	0.541	+0.004	14 of 20
Random for- est, depth 6, leaf 50	0.829	+0.055	0.556	−0.007	14 of 20
Random for- est, leaf 200	0.805	+0.047	0.550	−0.021	14 of 20
Penalised lo- gis- tic re- gres- sion	0.741	+0.045	0.510	−0.024	14 of 20

ment would be this paper’s own result in miniature rather than a finding about thermal dryness. The arm was therefore run here rather than deferred.

Both channels were dropped and everything refitted, over all twenty directions and all five within-region folds. The reference configuration reproduces the frozen exports exactly, at 0.000000 against both the transfer and the within-region references.

Table A.18:

Region	Increment, full set	Increment without the two channels	Retained
Manavgat 2021	+0.067	+0.063	94 %
Bejís 2022	+0.056	+0.046	82 %
Muğla 2021	+0.116	+0.097	84 %
North Evia 2021	+0.153	+0.145	94 %
Montiferru 2021	+0.101	+0.105	103 %
Mean	+0.099	+0.091	92 %

The increment does not depend on them. It stays positive in every region and retains 82 % to 103 % of its size, rising slightly in one region. Mean transfer is likewise unchanged, at 0.545 without the two channels against 0.541 with them and 0.537 for the baseline alone. The local skill this paper reports is therefore not an artefact of a coordinate-smoothed surface.

This arm was computed independently here and reproduces the companion paper’s figure for the same quantity, 82 % to 103 %, from a separately written harness.

Appendix A.1. A(i). The four evaluations of Section 4.3, in full

Section 4.3 reports four evaluations of the same models as a ladder. The
1650 per-split and per-scar detail is here, so that the ladder can be checked without
leaving the manuscript.

Table A1. Within-region half-split, every split. Source and target
positive counts are given because they are unequal, which is the principal
limit on this arm: a straight cut does not produce two exchangeable halves.
1655 Two splits are unusable because one half of Manavgat contains no burned
cells.

Table A.19:

Region	Axis	Direction	Source positives	Target positives	Thermal AUC	Baseline AUC
manavgat 2021	east-west	low to high	700	84	0.695	0.661
manavgat 2021	east-west	high to low	84	700	0.557	0.580
manavgat 2021	north-south	low to high	0	784	—	—
manavgat 2021	north-south	high to low	784	0	—	—
bejis 2022	east-west	low to high	640	460	0.750	0.784
bejis 2022	east-west	high to low	460	640	0.578	0.560
bejis 2022	north-south	low to high	357	743	0.713	0.662
bejis 2022	north-south	high to low	743	357	0.603	0.613
mugla 2021	east-west	low to high	1,449	1,462	0.448	0.378
mugla 2021	east-west	high to low	1,462	1,449	0.636	0.628
mugla 2021	north-south	low to high	774	2,137	0.533	0.503
mugla 2021	north-south	high to low	2,137	774	0.294	0.549
evia 2021 extended	east-west	low to high	2,052	612	0.589	0.506
evia 2021 extended	east-west	high to low	612	2,052	0.645	0.482
evia 2021 extended	north-south	low to high	2,564	100	0.649	0.513
evia 2021 extended	north-south	high to low	100	2,564	0.533	0.495
montiferru 2021	east-west	low to high	410	129	0.475	0.461
montiferru 2021	east-west	high to low	129	410	0.507	0.513
montiferru 2021	north-south	low to high	438	101	0.583	0.511
montiferru 2021	north-south	high to low	101	438	0.536	0.447

Table A2. Leave-one-scar-out at a 2 km buffer, every scar. Muğla

and Montiferru are the only regions containing more than one burned component of at least 50 cells. Muğla is the only region where holding one out still
1660 leaves the source model properly trained across every arm; Montiferru’s two components are very unequal, so one of its arms retains 472 source positives and the other 97. Muğla’s four arms mean 0.579; the four arms in the other regions mean 0.525, and the pooled figure of 0.552 averages the two.

Table A.20:

Region	Component	Source positives left	Target positives	Target cells	AUC
evia 2021 extended	1	11	2,653	3,059	0.465
manavgat 2021	1	88	696	1,135	0.592
montiferru 2021	1	97	442	758	0.584
montiferru 2021	5	472	67	195	0.458
mugla 2021	6	1,997	914	1,266	0.595
mugla 2021	1	2,173	738	1,244	0.561
mugla 2021	10	2,272	639	940	0.617
mugla 2021	8	2,363	548	954	0.542

The evaluation populations of the two arms are not comparable with
1665 each other or with the transfer targets. A held-out scar with its 2 km collar contains only fire-adjacent negatives, against a whole target region’s inclusion of its easy far field; the burned fractions, 34 to 87 % against 3.8 to 28.7 %, are a symptom of that rather than the cause, since ROC-AUC is invariant to class balance at fixed class-conditional distributions. The scar arm therefore
1670 asks for discrimination against the nearest and most similar negatives only, while a transfer arm includes the whole easy far field. Section 4.3 states the

consequence: the last three rows of the ladder are not distinguishable by this design.

Table A3. The foreign-region arm, decomposed by source. Each
1675 held-out scar area is scored with a model fitted on each of the other four regions. Table 4’s row D is the mean over the **eight** scars that carry a row C, that is 32 of the 36 combinations below; the nine-scar mean quoted in this appendix is 0.559 against row D’s 0.555.

Table A.21:

Target region	Scar	Mean over sources	Min	Max	Spread
Bejís 2022	1	0.589	0.521	0.631	0.110
North Evia 2021	1	0.590	0.374	0.704	0.329
Manavgat 2021	1	0.468	0.407	0.563	0.156
Montiferru 2021	1	0.504	0.457	0.546	0.089
Montiferru 2021	5	0.490	0.399	0.656	0.257
Muğla 2021	1	0.518	0.425	0.608	0.183
Muğla 2021	6	0.568	0.533	0.597	0.064
Muğla 2021	8	0.632	0.570	0.724	0.155
Muğla 2021	10	0.671	0.579	0.716	0.137

Over all 36 combinations here the mean is 0.559, the range 0.374 to 0.724,
1680 and ten fall below chance. Restricted to the 32 combinations behind row D the mean is 0.555 and the same ten fall below chance. The mean spread across the four sources for a single scar is 0.164 over the nine scars and 0.171 over the eight. Averaging over sources is what makes row D comparable with row C, which is fitted on one region; it is not a claim that the choice of
1685 foreign source is immaterial, and Section 4.3 states the distinction.

Table A4. Prevalence is not the cause of the evaluation-area effect. The same fitted model and the same out-of-fold predictions are scored three ways: on the whole region, on a random sample of region cells drawn at the scar area’s own burned fraction, and on the scar area. Twenty draws
1690 per scar, seed 42. Nine scars, since this control needs no leave-one-scar-out arm and Bejís therefore qualifies.

Table A.22:

Target region	Scar	Burned fraction	A whole region	A' B scar area prevalence- matched
Manavgat 2021	1	0.61	0.797	0.796 0.589
Bejís 2022	1	0.66	0.824	0.825 0.575
Muğla 2021	6	0.72	0.777	0.772 0.564
Muğla 2021	1	0.59	0.777	0.779 0.654
Muğla 2021	10	0.68	0.777	0.779 0.673
Muğla 2021	8	0.57	0.777	0.782 0.761
North Evia 2021	1	0.87	0.864	0.866 0.745
Montiferru 2021	1	0.58	0.720	0.717 0.622
Montiferru 2021	5	0.34	0.720	0.723 0.461
Mean			0.782	0.782 0.627

A minus A', the effect of prevalence alone, is **−0.000** [−0.003, +0.002]. A' minus B, the effect of replacing the region’s negatives with fire-adjacent ones, is **+0.155** [+0.093, +0.217]. The evaluation-area effect is entirely
1695 the negative pool.

Appendix B. Appendix B. All twenty transferability diagnostics

Section 4.5 summarises this table by family. The full ranking is given here because the negative result is the point: a reader should be able to see every candidate that was tried, not only the families' best members, and to
1700 check that no diagnostic was dropped after it failed.

Table B1. All transferability diagnostics versus raw thermal transfer (20 ordered directions). Spearman ρ with pair-based bootstrap 95 % CIs. Exp. = expected sign. Rows with $n = 12$ exist only for the four-AOI subset, because those diagnostics were never produced for Montiferru.
1705 The supported-features conditional rows use the 16 directions (8 pairs) with at least one CI-supported feature.

Table note. A null row means the diagnostic was **not shown to order transfer** on this design, not that it was shown incapable of ordering it. With ten effective region pairs the power is low, and the intervals are wide
1710 enough to admit moderate true correlations in either direction. The last row is retained rather than deleted because it was computed: on two directions the statistic has no meaningful value, and reporting that is more honest than dropping the variant.

Reading the two rows that clear zero. Both are supported-feature
1715 variants, where the predictor subset is chosen by whether two regions' bootstrap intervals happen to be disjoint. That is a data-dependent selection made on the same data, uncorrected. Their all-nine-feature counterparts are in the table and both span zero. The result lives in the selection step, and Section 4.5 says so.

1720 **Equal-sample check.** The families sit on unequal samples: marginal,

Table B.23:

Diagnostic	Family	Exp.	n dir	Spearman ρ [95% CI]	CI excludes 0
Agreement fraction, supported features	$P(y x)$ conditional	+	16	+0.84 [+0.58, +0.88]	yes
Cosine, supported features	$P(y x)$ conditional	+	16	+0.81 [+0.33, +0.88]	yes
Cosine, all 9 features	$P(y x)$ conditional	+	20	+0.50 [−0.17, +0.83]	no
Vector Spearman, all 9	$P(y x)$ conditional	+	20	+0.27 [−0.36, +0.77]	no
Agreement count, all 9	$P(y x)$ conditional	+	20	+0.18 [−0.40, +0.72]	no
Schoener’s D, 1-D mean	$P(x y=1)$ niche	+	20	+0.24 [−0.45, +0.74]	no
Warren’s I, 1-D mean	$P(x y=1)$ niche	+	20	+0.22 [−0.42, +0.73]	no
Schoener’s D, PCA-2D	$P(x y=1)$ niche	+	20	+0.10 [−0.51, +0.68]	no
Warren’s I, PCA-2D	$P(x y=1)$ niche	+	20	−0.07 [−0.66, +0.49]	no
Mahalanobis, burned centroids	$P(x y=1)$ niche	−	20	−0.23 [−0.75, +0.44]	no
Domain-classifier AUC	$P(x)$ marginal	−	20	−0.32 [−0.78, +0.33]	no
Predictor-space mean dissimilarity	$P(x)$ marginal	−	12	−0.10 [−0.54, +0.43]	no
Predictor-space p95 dissimilarity	$P(x)$ marginal	−	12	−0.08 [−0.59, +0.49]	no
Fraction inside weighted AoA	$P(x)$ marginal	+	12	+0.22 [−0.48, +0.59]	no
Fraction inside	$P(x)$ marginal	+	12	+0.08 [−0.89, +0.43]	no

applicability, climatic and geographic rows on twelve directions, the supported-
conditional rows on sixteen, the rest on twenty. Recomputing every row on
the common twelve directions reproduces the published values to 4.8×10^{-5}
and leaves the ordering unchanged — the conditional rows still lead at +0.87
1725 [+0.65, +0.88] and +0.85 [+0.43, +0.88], every marginal row still spans zero.
Source: `paper/diagnostics_common_subset.md`.

*Appendix B.1. B2. The signed univariate associations the reversal claim
rests on*

Section 4.7b drops two predictors because their signed association with
1730 burning reverses between regions with bootstrap support on the frames as
drawn. Section 4.10 shows that both supported elevation reversals disappear
once the frames are equalised, so this table is the evidence for the feature-
removal arm and for the narrowed mechanism claim inside Contribution 1,
not for a general reversal mechanism. It was computed for the frozen analysis
1735 and is reproduced here because the claim is otherwise asserted rather than
shown.

**Table B2. Signed univariate AUC of each predictor against
burned, by region.** Primary natural-vegetation population; 10-cell (~ 5
km) spatial-block bootstrap, 1000 replicates, seed 42. The AUC is never
1740 folded to $\max(\text{AUC}, 1 - \text{AUC})$, so a value below 0.5 means lower values
rank burned and is a direction rather than weakness. **Bold** marks a region
whose own interval excludes 0.5.

What counts as a reversal. A pair of regions is called a reversal
only when their signed associations point to opposite sides of 0.5 **and each**
1745 **region’s own interval excludes 0.5**. That is stricter than requiring the

Table B.24:

Feature	Manavgat	Bejís	Muğla	Evia	Montiferru
elevation_	0.374 [0.289,	0.643 [0.558,	0.611 [0.532,	0.541 [0.448,	0.584 [0.395,
mean	0.471]	0.729]	0.690]	0.626]	0.762]
slope_mean	0.531 [0.423,	0.521 [0.439,	0.637 [0.582,	0.487 [0.418,	0.652 [0.506,
	0.642]	0.605]	0.686]	0.554]	0.771]
ndvi_mean	0.636 [0.587,	0.559 [0.497,	0.662 [0.616,	0.639 [0.575,	0.586 [0.450,
	0.676]	0.619]	0.704]	0.701]	0.704]
lst_anomaly_	0.482 [0.428,	0.418 [0.364,	0.485 [0.395,	0.640 [0.567,	0.395 [0.285,
mean	0.530]	0.480]	0.566]	0.710]	0.535]
current_lst_	0.538 [0.452,	0.477 [0.401,	0.325 [0.271,	0.377 [0.301,	0.370 [0.248,
mean	0.621]	0.547]	0.382]	0.456]	0.513]
current_	0.552 [0.460,	0.517 [0.429,	0.336 [0.275,	0.362 [0.285,	0.356 [0.233,
tvdi_mean	0.641]	0.595]	0.398]	0.442]	0.499]
tvdi_	0.449 [0.391,	0.512 [0.443,	0.490 [0.396,	0.519 [0.444,	0.378 [0.282,
difference_	0.505]	0.583]	0.575]	0.589]	0.497]
mean					
downscaled_	0.552 [0.466,	0.484 [0.400,	0.307 [0.253,	0.377 [0.297,	0.365 [0.240,
lst_mean	0.637]	0.560]	0.366]	0.459]	0.511]
fused_lst_	0.540 [0.454,	0.481 [0.404,	0.325 [0.272,	0.376 [0.300,	0.370 [0.248,
mean	0.622]	0.551]	0.383]	0.456]	0.513]

two regions' intervals to be disjoint, and the difference matters: for `current_`
`lst_mean` between Manavgat and Muğla the two intervals are disjoint, but
Manavgat's own interval, $[0.452, 0.621]$, includes 0.5, so Manavgat has no
established direction to reverse from. That pair is a point reversal, not a
1750 supported one.

Table B3. The cross-region reversals that meet the stricter criterion. Difference intervals are from the same paired bootstrap.

Table B.25:					
Feature	Region A	AUC	Region B	AUC	Difference [95 % CI]
<code>elevation_mean</code>	Manavgat	0.374	Bejís	0.643	+0.269 [+0.138, +0.397]
<code>elevation_mean</code>	Manavgat	0.374	Muğla	0.611	+0.235 [+0.102, +0.360]
<code>lst_anomaly_mean</code>	Bejís	0.418	Evia	0.640	+0.221 [+0.123, +0.313]

Three pair-level reversals across **two** features, which is why Section 3.12
removes exactly those two. Twenty-nine further pairs reverse at the point
1755 estimate only, spread across eight of the nine features, and they are not
counted. The conservative criterion costs the paper findings rather than
manufacturing them: a difference interval on the pair, which is the instrument
Section 4.9 uses, would support more reversals than the three listed here.

Appendix B.2. B4. The transfer matrix in precision-recall terms

1760 ROC-AUC is reported throughout the main text for comparability with the susceptibility literature. A susceptibility surface is used as a ranked area budget, so precision-recall is the operational quantity, and at target prevalences of 3.8 to 28.7 % the two can differ sharply. Read from the same frozen step9b exports as Table 7.

1765 **Table B4. Thermal transfer, PR-AUC against the no-skill baseline.** The baseline is the target’s burned prevalence. Lift is PR-AUC divided by that baseline; a lift of 1 is no better than random ranking. Ordered by lift.

Six directions fall below their own no-skill baseline, and only one exceeds
1770 twice it. The six are the same six that are below chance on ROC-AUC, which is what a reversed ranking predicts in either metric. Section 4.10 shows that this count is largely a property of the evaluation frames rather than of a reversed predictor-burning relationship.

Appendix C. Appendix C. Protocol detail

1775 These protocols are given here in full rather than in Methods, because each is a specification a reader needs only when checking the corresponding result, and none is needed to follow the argument. Sections 3.2, 3.3, 3.11 and 3.15 state in summary what each does.

*Appendix C.1. C.1 Burned-area label, the reconstructed analysis grid, and
1780 the admissibility gate*

Labels come from MODIS MCD64A1 Collection 6.1 (MODIS/061/MCD64A1) retrieved through Google Earth Engine [11], whose omission and commission

Table B.26:

Direction	ROC-AUC	PR-AUC	No-skill	Lift
Evia $\rightarrow$ Manavgat	0.686	0.094	0.038	2.45
Bejís $\rightarrow$ Montiferru	0.594	0.289	0.212	1.36
Evia $\rightarrow$ Montiferru	0.647	0.283	0.212	1.34
Bejís $\rightarrow$ Muğla	0.619	0.093	0.070	1.33
Muğla $\rightarrow$ Evia	0.653	0.379	0.287	1.32
Montiferru $\rightarrow$ Bejís	0.548	0.093	0.072	1.28
Montiferru $\rightarrow$ Muğla	0.619	0.089	0.070	1.27
Evia $\rightarrow$ Muğla	0.577	0.086	0.070	1.23
Muğla $\rightarrow$ Bejís	0.583	0.088	0.072	1.22
Manavgat $\rightarrow$ Evia	0.613	0.343	0.287	1.20
Manavgat $\rightarrow$ Montiferru	0.533	0.243	0.212	1.15
Montiferru $\rightarrow$ Manavgat	0.567	0.043	0.038	1.11
Montiferru $\rightarrow$ Evia	0.586	0.317	0.287	1.11
Muğla $\rightarrow$ Montiferru	0.531	0.214	0.212	1.01
Bejís $\rightarrow$ Manavgat	0.444	0.034	0.038	0.90
Manavgat $\rightarrow$ Muğla	0.470	0.063	0.070	0.90
Evia $\rightarrow$ Bejís	0.448	0.059	0.072	0.82
Bejís $\rightarrow$ Evia	0.383	0.226	0.287	0.79
Muğla $\rightarrow$ Manavgat	0.401	0.029	0.038	0.75
Manavgat $\rightarrow$ Bejís	0.326	0.049	0.072	0.67
Mean	0.541	0.156	0.136	1.16

characteristics [4] bound every model here. The product is exported onto the 30 m EPSG:4326 reference grid used by the rest of the pipeline, which
1785 duplicates each native ~ 500 m observation across the 30 m pixels beneath it.

The analysis grid is therefore **reconstructed rather than native**: the 30 m grid is partitioned into non-overlapping blocks of $\text{round}(500 / 30) = 17 \times 17$ pixels, giving a nominal cell edge of 510 m, and each cell is identified
1790 by its integer block indices `row_500m` and `col_500m`. This approximates the MODIS sinusoidal cell rather than reproducing it, being anchored to the Landsat/EPSG:4326 reference grid, and marginal blocks are truncated. The cell is square in degrees but not on the ground, about 510 m north to south and 390 to 407 m east to west, so its area is 0.199 to 0.208 km² against a
1795 MODIS cell's 0.250 km²; every block-size label in this paper is therefore the north-south dimension. The companion paper develops what follows from that.

A cell's representative burn date is the **mode** of the positive sub-pixel day-of-year values in the block, tested against the region's label window. Be-
1800 cause the exported raster carries no out-of-window positives, in this dataset a single in-window positive sub-pixel makes the cell burned. The fraction of positive sub-pixels agreeing with the modal date is recorded as `burn_date_pixel_agreement_fraction` but no agreement threshold is imposed. The label never affects a cell's eligibility for modelling: unburned, all-no-data
1805 and out-of-window cells all remain as the negative class.

Two safeguards are recorded rather than assumed. Cells that burned inside a region's own predictor window are removed from its analysis universe

rather than counted as unburned, which ran for Muğla, Evia and Montiferru, is recorded as not run for Manavgat and has no recorded status for Bejís.

1810 Burning in earlier years is screened for no region in the five-region cohort; the only historical exclusion in the study removes the 2021 Muğla scar from the 2022 event-relative experiment of Section 3.15.

Before any modelling each region passes a gate that answers one question: of the cells labelled burned, what fraction is dominated by natural

1815 vegetation? A region is admitted as a wildfire candidate when that fraction reaches 0.50 and at least 30 burned cells are present, and is rejected as a cropland-dominated control when the cropland fraction reaches 0.50 instead. The gate uses ESA WorldCover classes aggregated to the same reconstructed cells. Its purpose is to separate burned area produced by natural-fuel combustion from burned area produced by post-harvest stubble burning, which

1820 MCD64A1 does not distinguish. Verdicts are reported in Section 4.1.

Appendix C.2. C.2 Transferability diagnostics versus transfer

Twenty candidate diagnostics from five families are each rank-correlated against the same target quantity, the raw thermal transfer AUC over the

1825 twenty ordered directions, under one common pair-based bootstrap. The families are marginal predictor-distribution measures $P(x)$, burned-niche overlap $P(x|y=1)$, fire-regime label-pattern structure $P(y)$, and conditional feature-response direction $P(y|x)$.

The **marginal** family includes area-of-applicability-style dissimilarity in scaled, importance-weighted predictor space [23, 24, 18], climatic and geographic distance, and a learned domain classifier trained to separate source

1830 from target cells. The **niche-overlap** family includes Schoener’s D [36] and

Warren’s I [44] over the burned cells of each region. The **regime** family includes burned-patch component counts and effective component counts. The
1835 **conditional** family is the sign-agreement index: the fraction of predictors whose signed univariate association points the same way in both regions, computed over all predictors and over the subset whose associations are individually interval-supported.

Two properties of this design are stated in advance because they bound
1840 what it can show. The candidate set was fixed in a project analysis log before the correlations were computed; that log is not a formal pre-registration and no independent timestamped registration exists. And with five regions the effective sample is ten unordered pairs, so nineteen computed variants are evaluated at that power with no family-wise error control claimed; where a
1845 diagnostic clears an interval test but not a Bonferroni threshold, both are reported.

Appendix C.3. C.3 Same-geography event-to-event comparison (Muğla 2021 versus 2022)

Muğla is the one region for which a second fire event is analysed on the
1850 identical AOI and analysis grid, which allows the direction of feature-label associations to be compared with geography held fixed. The 2022 experiment is registered with windows anchored to its own event: a 58-day predictor window closing the day before ignition and a 49-day label window opening on it, matching the 2021 durations exactly.

This is not a clean temporal-transfer design and is not presented
1855 **as one.** The 2022 event ignites about six weeks earlier in the season than the 2021 event, so calendar year and seasonal phase are confounded and no

difference can be attributed to the year alone. The registered claim is same-
geography event-to-event. The pair is deliberately kept out of the twenty-
1860 direction matrix, and that exclusion is enforced in released code rather than
merely asserted here.

Two further properties bound the comparison. A historical-burn exclusion
applies to the 2022 arm alone, masking every cell that burned in 2021 so the
previous year’s scar does not enter the following year’s analysis: it removes
1865 3,073 cells, of which 2,941 belong to the primary population, taking it from
41,730 rows to 38,790, a drop of 7.0 %. The two arms therefore share 38,789 of
38,790 cells and three byte-identical static predictors, and in the 2022-to-2021
direction every target positive lies outside the source training population.
Section 4.9 states what follows for how these two numbers may be read.

1870 **CRediT authorship contribution statement**

Yunus Emre Cogurcu: Conceptualization, Methodology, Formal anal-
ysis, Investigation, Writing – original draft, Writing – review and editing,
Supervision. **Emrehan Metin:** Software, Data curation, Investigation, Val-
idation, Writing – review and editing.

1875 **Declaration of competing interest**

The authors declare that they have no known competing financial inter-
ests or personal relationships that could have appeared to influence the work
reported in this paper.

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Data availability

1885 All satellite inputs are public and are obtained through Google Earth Engine. The processing pipeline, its configuration and the frozen numeric outputs are publicly available; the repository, the commit of record and the licence are given in the data and code availability statement of the Methods, together with the three components that fall outside that release. No digital object identifier is minted and no archival deposit exists.

References

- 1890 [1] Alkan Akıncı, H., Akıncı, H., 2023. Machine learning based forest fire susceptibility assessment of Manavgat district (Antalya), Turkey. *Earth Science Informatics* 16, 397–414. doi:10.1007/s12145-023-00953-5.
- [2] Archibald, S., Lehmann, C.E.R., Gómez-Dans, J.L., Bradstock, R.A., 2013. Defining pyromes and global syndromes of fire regimes. *Proceedings of the National Academy of Sciences* 110, 6442–6447. doi:10.1073/pnas.1211466110.
- 1895 [3] Barve, N., Barve, V., Jiménez-Valverde, A., Lira-Noriega, A., Maher, S.P., Peterson, A.T., Soberón, J., Villalobos, F., 2011. The crucial role of the accessible area in ecological niche modeling and

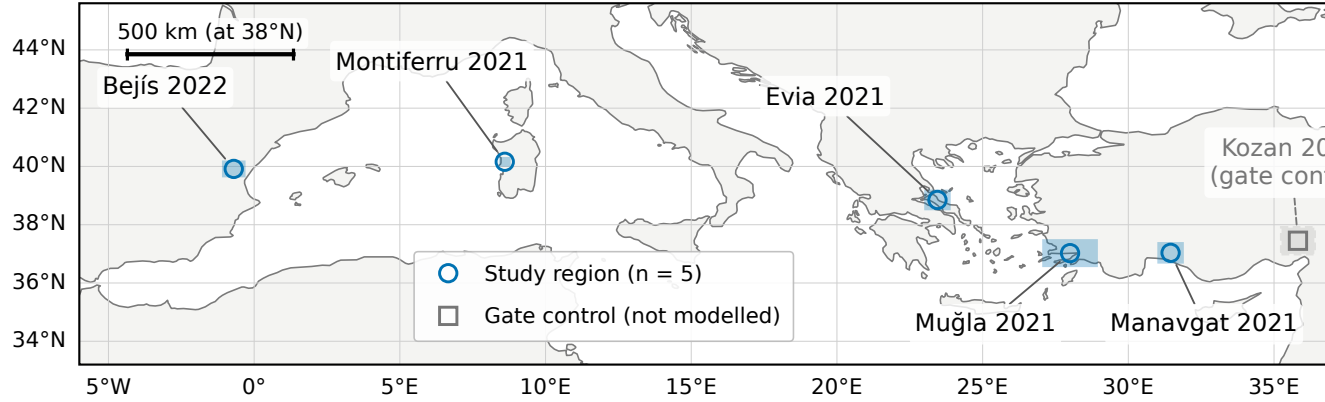


Figure C.1: **Study regions and the gate negative control.** Locations of the five wild-fire regions analysed in this study (blue, circular markers): Manavgat 2021 and Muğla 2021 (Türkiye), Bejís 2022 (Spain), North Evia 2021 (Greece, extended AOI) and Montiferru 2021 (Sardinia, Italy). Rectangles are the true analysis areas of interest; the ring marks each centroid, because at basin scale the rectangles alone are hard to locate. Kozan 2023 (grey, dashed outline, square marker) is *not* a study region: it is the negative control used to calibrate the burned-landcover admissibility gate, and it enters no model, transfer or diagnostic anywhere in this paper. Its rectangle is the true extent of its processed AOI and is larger than the study AOIs for that reason alone; the difference in size carries no methodological meaning and is not a design choice. Coastlines are Natural Earth 1:50 m; country borders are omitted deliberately. The scale bar is exact only at 38°N. Region bounding boxes match Table 1 to 5×10^{-3} degrees.

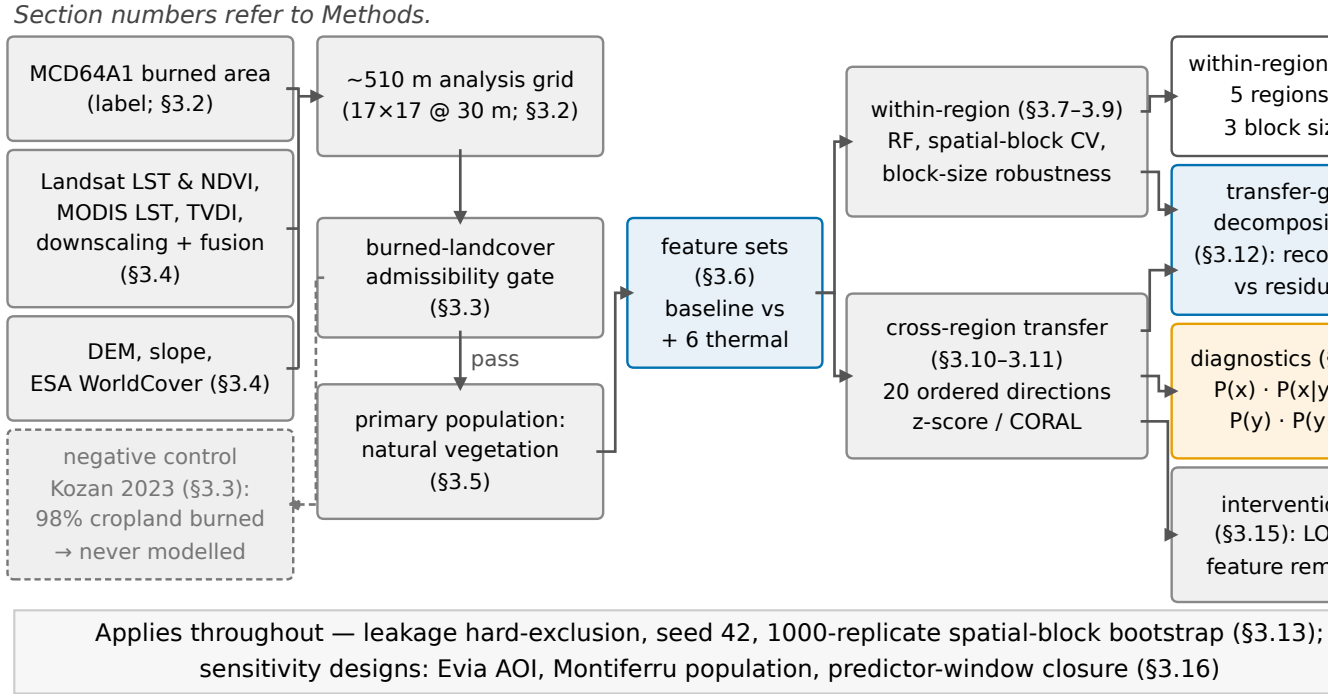


Figure C.2: Processing pipeline and evaluation programme. Flow from source products (left) to the transfer and diagnostic programme (right); every box carries the Methods subsection that specifies it. The five source products (MCD64A1 burned area, Landsat Collection 2 Level-2, MOD11A1, Copernicus DEM GLO-30 and ESA WorldCover) merge onto the reconstructed ~ 510 m analysis grid (17×17 pixels of the 30 m reference grid). The burned-landcover admissibility gate has two outcomes: regions that *pass* proceed to the natural-vegetation primary population, while the rejecting branch (grey, dashed) terminates in the Kozan 2023 negative control, whose burned area is 98% cropland and which is never modelled. The dashed connector deliberately carries no arrowhead onward, so the control cannot be read as entering the pipeline. Feature sets feed two evaluations in parallel: within-region (random forest under spatial-block cross-validation) and cross-region transfer (source-only, 20 ordered directions, with label-blind z-score and CORAL adaptation). Their levels are combined in the transfer-gap decomposition (§3.12). The footer states what applies throughout: leakage hard-exclusion, seed 42, 1000-replicate spatial-block bootstrap (§3.13), and the sensitivity designs (§3.16). Section references are verified against the Methods heading list when the figure is built.

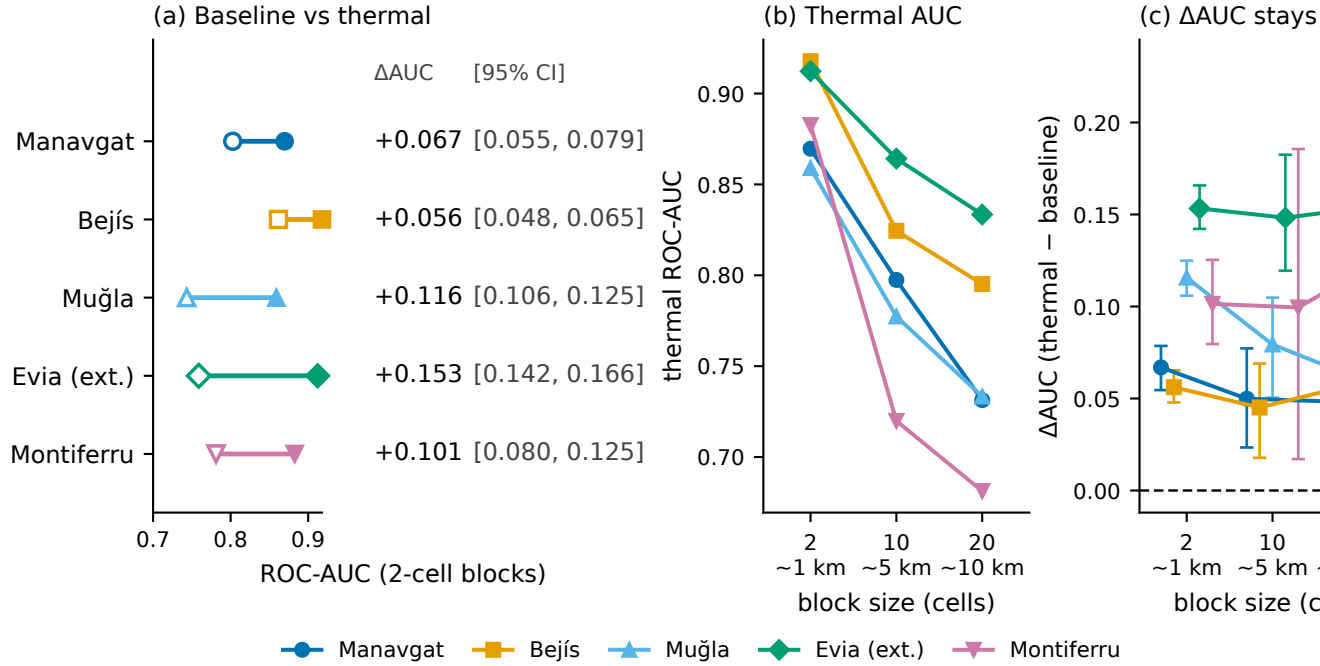


Figure C.3: **The thermal increment replicates in all five regions and survives coarser spatial blocking.** Natural-vegetation primary population; random forest with the primary weighted configuration; all intervals are 95% spatial-block bootstrap intervals (1000 replicates, seed 42). **(a)** Baseline versus thermal out-of-fold ROC-AUC per region at the default 2-cell (~ 1 km) blocking: the open marker is the static baseline feature set, the filled marker is the baseline plus the six thermal predictors, and the connecting bar is the increment. The right-hand column reports the paired ΔAUC and its 95% interval; it is a fixed column placed beyond the plotted AUC range, so no label position depends on a data value. Every interval excludes zero. **(b)** Absolute thermal ROC-AUC against spatial block size (2, 10 and 20 cells, ≈ 1 , 5 and 10 km): absolute skill falls as blocking coarsens, in every region. **(c)** The thermal increment ΔAUC against the same block sizes, with 95% intervals: the increment does *not* melt away, and its interval excludes zero in all ten of the 2- and 10-cell region-block-size combinations. The five 20-cell intervals also lie above zero but rest on 6 to 33 positive-carrying blocks each, so they are plotted as indicative rather than as intervals (Table 3, table note). In (c) the five regions are offset horizontally by 0.15 cell so their whiskers separate; **this offset is cosmetic, since all five regions are evaluated at exactly the same three block sizes** (2, 10, 20 cells), and the offset is smaller than half the spacing between block sizes, so the groups cannot merge. Colour and marker shape both encode region, so the panels remain readable in greyscale.

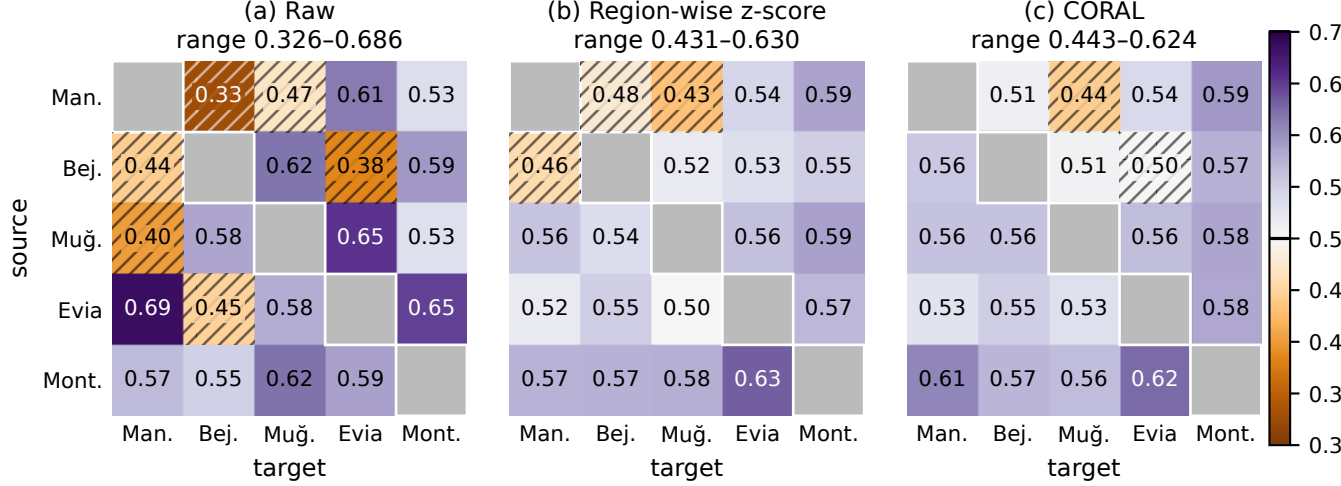


Figure C.4: **Cross-region transfer, and what label-blind adaptation does to it.** Thermal transfer ROC-AUC for all 20 ordered source–target directions among the five regions, natural-vegetation primary population. **(a)** Raw transfer, no adaptation. **(b)** After region-wise z-score standardisation. **(c)** After CORAL. Rows are the source region, columns the target; the grey diagonal is within-region performance and is *not* a transfer result. The diverging colour scale is centred exactly on chance (0.5, marked by the black rule on the colour bar), so orange is below chance and purple above. Each panel header states that panel’s range: adaptation compresses the spread from 0.326–0.686 (raw) to 0.431 to 0.630 (z-score) and 0.443 to 0.624 (CORAL). It pulls the extremes toward chance rather than lifting transfer. Cells below chance are hatched. The hatch is a redundant cue for print: a diverging palette is symmetric in luminance about its centre, so in greyscale a below-chance cell would otherwise be indistinguishable from an equally distant above-chance one. Hatching follows the *exact* value while the printed label is rounded to two decimals, so two cells print “0.50” with different hatching: CORAL Bejís→Evia is 0.4991 (hatched, below chance) and z-score Evia→Muğla is 0.5013 (un-hatched, above chance). Every cell carries its value, and every label’s colour is chosen by measured WCAG contrast against that cell’s rendered colour (worst case 4.7:1). Confidence intervals for these directions are given in Table 4.

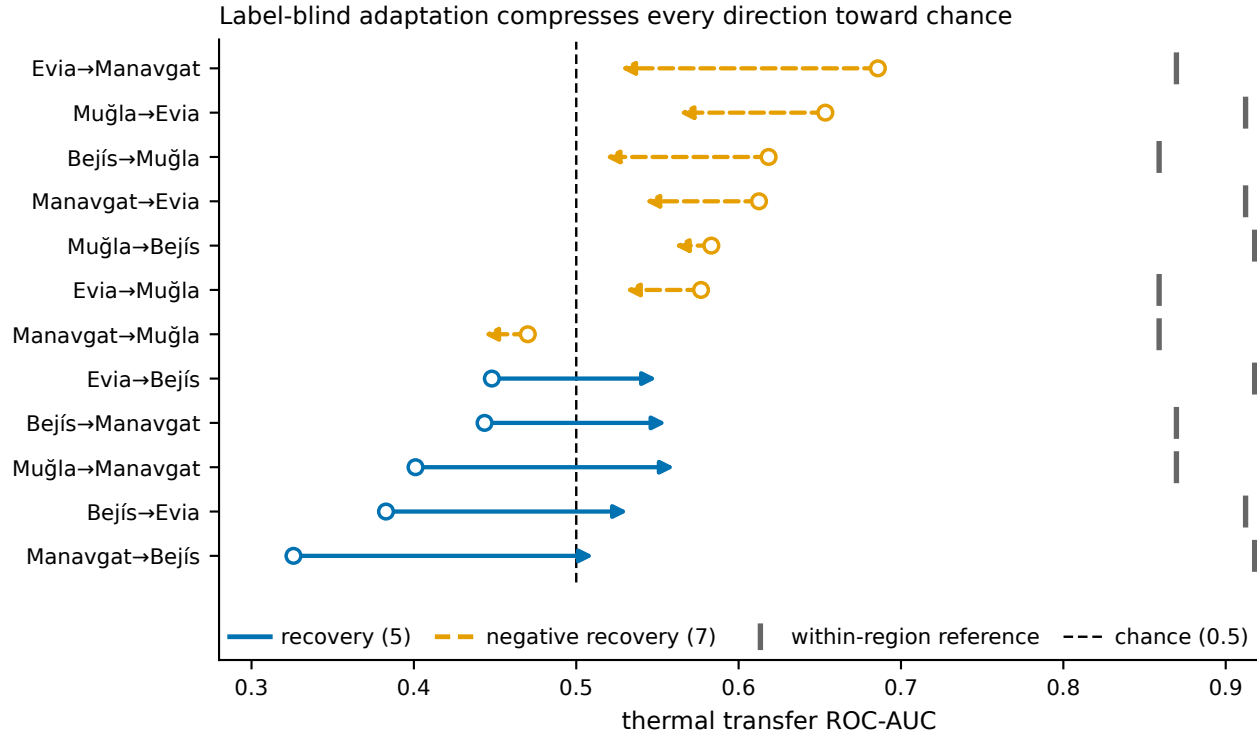


Figure C.5: **Label-blind adaptation regresses transfer onto the level a model reaches on an unseen scar, rather than recovering it.** All 12 ordered directions among the four regions carrying the full decomposition (Manavgat, Bejís, Muğla, Evia), natural-vegetation primary population. Each row is one direction. The open circle is raw transfer ROC-AUC and the arrow runs to the best label-blind adapted result for that direction (region-wise z-score or CORAL, whichever is better); the grey vertical tick is the within-region reference for that target, and the dashed vertical rule is chance. **Solid blue arrows are the five directions where adaptation moves transfer toward the within-region reference (recovery); dashed orange arrows are the seven where it moves *away* from it (negative recovery).** Line style and colour both encode this class, because the two hues differ by only 2.30:1 in relative luminance and would nearly merge in greyscale. The pattern is the point: all six directions starting above chance are pushed down, and five of the six starting below chance are pushed up, so adaptation acts on the spread rather than on skill. The exception is Manavgat→Muğla, which starts below chance at 0.470 and is pushed further down to 0.443. The worst case is Evia→Manavgat, where raw transfer of 0.686 is dragged down to 0.527 against a within-region reference of 0.870, a recovered fraction of -0.862 . No arrow reaches its within-region tick.

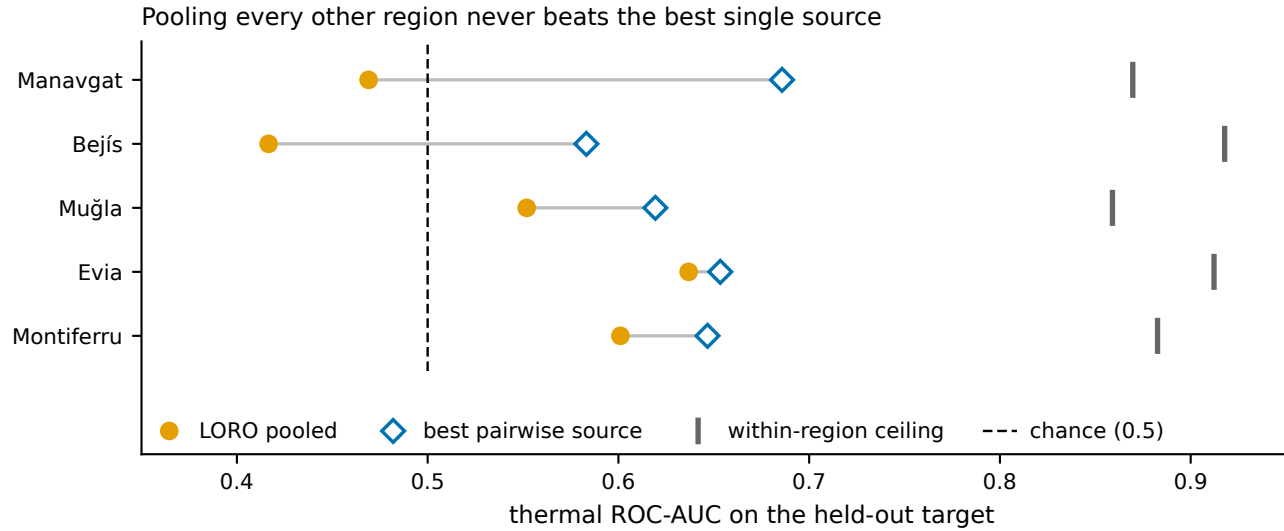


Figure C.6: **Pooling every other region never beats the best single source.** Leave-one-region-out evaluation: for each held-out target region (rows), a model trained on the pooled remaining four regions (filled orange circle) is compared against the best individual source region for that same target (open blue diamond), with the within-region ceiling as a grey vertical tick and chance as the dashed rule. Natural-vegetation primary population, thermal feature set, no adaptation. The pooled model lies to the left of the best pairwise source in all five targets without exception. Adding source regions does not accumulate transferable signal. In two targets the pooled model falls *below chance* at the point estimate (Manavgat 0.469, Bejís 0.417), and only for Bejís does the interval lie entirely below chance, while a single well-chosen source reaches 0.686 and 0.583 respectively. Every point remains far below the within-region ceiling.

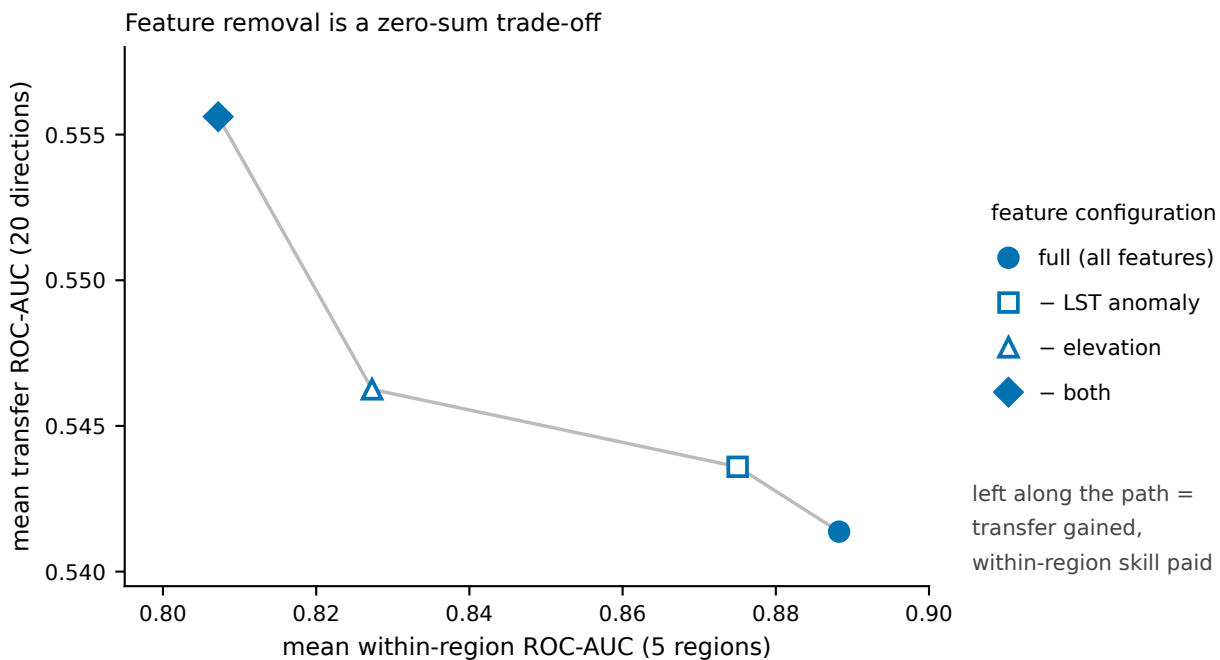


Figure C.7: **Removing the direction-reversing features costs locally and returns nothing measurable on transfer.** Mean within-region ROC-AUC across the five regions (horizontal) against mean transfer ROC-AUC across the 20 ordered directions (vertical), for four feature configurations: the full thermal set, the set without the LST anomaly, without elevation, and without both. Each configuration has its own marker and is named in the key; the grey path connects them in that order. Moving left along the path, transfer rises monotonically ($0.541 \rightarrow 0.556$) while within-region skill falls monotonically ($0.888 \rightarrow 0.807$). Both monotonicities are verified when the figure is built, so the figure cannot silently contradict the claim. The within-region cost is 0.081 AUC and is supported in every region. The change in transfer is +0.014 AUC on an interval that spans zero, so no exchange between the two is demonstrated: a local cost is measured and no compensating transfer gain is.

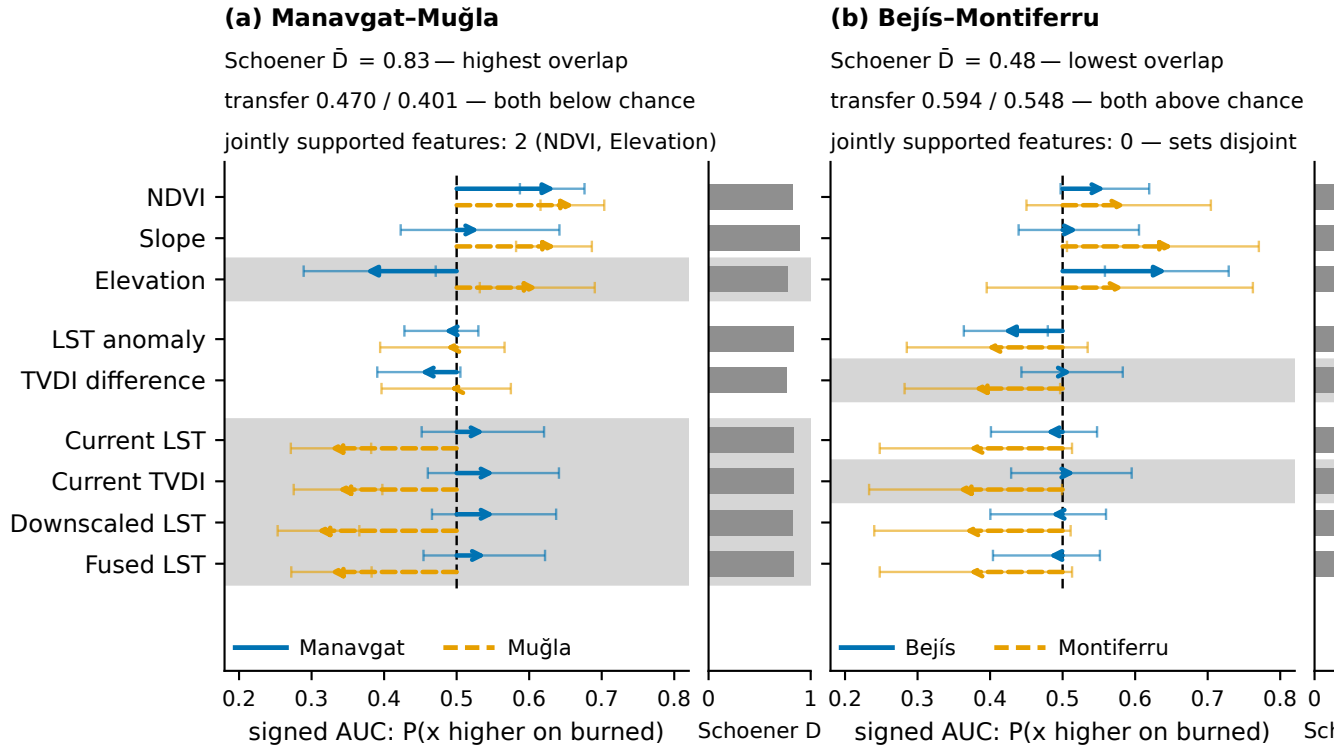


Figure C.8: **Burned-niche overlap does not determine transfer: the contrast pair.** Per-feature signed univariate AUC, the probability that a predictor takes a higher value on a burned cell than on an unburned one, for the two most informative region pairs, natural-vegetation primary population. Arrows run from chance (0.5) to the observed value, so their direction is the sign of the association and their length its strength; whiskers are 95% spatial-block (≈ 5 km) bootstrap intervals. Within each pair the two regions are distinguished by colour, by line style (solid for the first region named, dashed for the second) and by vertical offset; line style is included because the two hues differ by only 2.30:1 in relative luminance and would nearly merge in greyscale. **A filled arrowhead means that region's 95% interval excludes 0.5, so the direction is statistically supported; an open arrowhead means the interval contains 0.5 and the direction is not supported.** Rows shaded grey are features whose sign disagrees between the two regions. Grey side bars are the per-feature Schoener's D overlap. **(a)** Manavgat–Muğla, the pair with the *highest* burned-niche overlap ($\bar{D} = 0.83$), transfers below chance in both directions (0.470 and 0.401). **(b)** Bejís–Montiferru, the pair with the *lowest* overlap ($\bar{D} = 0.48$), transfers above chance in both directions (0.594 and 0.548). High niche overlap is therefore not sufficient for transfer. The arrowhead fill also exposes the coverage limit of the supported-agreement index, which is defined only on features whose direction is supported in *both* regions: there are two such features in (a) (NDVI and elevation) and **none in (b)**, where the two regions' supported sets are disjoint and Montiferru's intervals

- 1900 species distribution modeling. *Ecological Modelling* 222, 1810–1819.
doi:10.1016/j.ecolmodel.2011.02.011.
- [4] Boschetti, L., Roy, D.P., Giglio, L., Huang, H., Zubkova, M., Humber,
M.L., 2019. Global validation of the Collection 6 MODIS burned area
product. *Remote Sensing of Environment* 235, 111490. doi:10.1016/j.
1905 **rse.2019.111490.**
- [5] Breiman, L., 2001. Random forests. *Machine Learning* 45, 5–32. doi:10.
1023/A:1010933404324.
- [6] de Bruin, S., Brus, D.J., Heuvelink, G.B.M., van Ebbenhorst Tengber-
gen, T., Wadoux, A.M.J.C., 2022. Dealing with clustered samples for
1910 assessing map accuracy by cross-validation. *Ecological Informatics* 69,
101665. doi:10.1016/j.ecoinf.2022.101665.
- [7] Chuvieco, E., Cocero, D., Riaño, D., Martin, P., Martínez-Vega, J., de la
Riva, J., Pérez, F., 2004. Combining NDVI and surface temperature for
the estimation of live fuel moisture content in forest fire danger rating.
1915 *Remote Sensing of Environment* 92, 322–331. doi:10.1016/j.**rse.2004.**
01.019.
- [8] Dimarco, N.A., Faraji, I., Wahbi, M., Maatouk, M., Boulaassal, H.,
Yazidi Aalaoui, O., El Kharki, O., 2026. Generalizing human-driven
wildfire ignition models across Mediterranean regions using harmonized
1920 remote-sensing and machine-learning data. *Geomatics* 6, 13. doi:10.
3390/geomatics6010013.

- [9] Gelabert, P.J., Jiménez-Ruano, A., Ochoa, C., Alcasena, F., Sjöström, J., Marrs, C., Ribeiro, L.M., Palaiologou, P., Bentu Martínez, C., Chuvieco, E., Vega-García, C., Rodrigues, M., 2025. Assessing human-caused wildfire ignition likelihood across Europe. *Natural Hazards and Earth System Sciences* 25, 4713–4729. doi:10.5194/nhess-25-4713-2025.
- [10] Giglio, L., Boschetti, L., Roy, D.P., Humber, M.L., Justice, C.O., 2018. The Collection 6 MODIS burned area mapping algorithm and product. *Remote Sensing of Environment* 217, 72–85. doi:10.1016/j.rse.2018.08.005.
- [11] Gorelick, N., Hancher, M., Dixon, M., Ilyushchenko, S., Thau, D., Moore, R., 2017. Google Earth Engine: Planetary-scale geospatial analysis for everyone. *Remote Sensing of Environment* 202, 18–27. doi:10.1016/j.rse.2017.06.031.
- [12] Huang, Y., Liu, N., Chen, W., Tan, W., Deng, Y., Wang, Z., Chlus, A., Shen, J., Townsend, P.A., 2026. Enhancing transferability of foliar trait retrieval models: A comparative analysis of transfer learning strategies and domain shift characterization. *Remote Sensing of Environment* 337, 115336. doi:10.1016/j.rse.2026.115336.
- [13] Iban, M.C., Sekertekin, A., 2022. Machine learning based wildfire susceptibility mapping using remotely sensed fire data and GIS: A case study of Adana and Mersin provinces, Turkey. *Ecological Informatics* 69, 101647. doi:10.1016/j.ecoinf.2022.101647.

- 1945 [14] Jodhani, K., Bhatiya, L., Sirvi, L., Chanda, A., Thakkar, S., Rathnayake, U., 2026. Artificial intelligence and remote sensing frameworks for wildfire monitoring and risk analysis across multiple ecosystems: A review. *AIMS Environmental Science* 13, 56–98. doi:10.3934/environsci.2026004.
- 1950 [15] Kondylatos, S., Prapas, I., Camps-Valls, G., Papoutsis, I., 2023. Mesogeos: A multi-purpose dataset for data-driven wildfire modeling in the Mediterranean, in: *Advances in Neural Information Processing Systems* 36 (NeurIPS 2023) Datasets and Benchmarks Track, pp. 50661–50676. doi:10.52202/075280–2204.
- 1955 [16] Liu, C., Mostafavi, A., 2025. WildfireGenome: Interpretable machine learning reveals local drivers of wildfire risk and their cross-county variation. *arXiv preprint arXiv:2511.11589*. doi:10.48550/arXiv.2511.11589, arXiv:2511.11589.
- 1960 [17] Lobo, J.M., Jiménez-Valverde, A., Real, R., 2008. AUC: a misleading measure of the performance of predictive distribution models. *Global Ecology and Biogeography* 17, 145–151. doi:10.1111/j.1466-8238.2007.00358.x.
- [18] Ludwig, M., Moreno-Martinez, A., Hölzel, N., Pebesma, E., Meyer, H., 2023. Assessing and improving the transferability of current global spatial prediction models. *Global Ecology and Biogeography* 32, 356–368. doi:10.1111/geb.13635.
- 1965 [19] Maffei, C., Alfieri, S.M., Menenti, M., 2018. Relating spatiotemporal

- patterns of forest fires burned area and duration to diurnal land surface temperature anomalies. *Remote Sensing* 10, 1777. doi:10.3390/rs10111777.
- 1970 [20] Maffei, C., Lindenbergh, R., Menenti, M., 2021. Combining multispectral and thermal remote sensing to predict forest fire characteristics. *ISPRS Journal of Photogrammetry and Remote Sensing* 181, 400–412. doi:10.1016/j.isprsjprs.2021.09.016.
- 1975 [21] Maffei, C., Menenti, M., 2019. Predicting forest fires burned area and rate of spread from pre-fire multispectral satellite measurements. *ISPRS Journal of Photogrammetry and Remote Sensing* 158, 263–278. doi:10.1016/j.isprsjprs.2019.10.013.
- 1980 [22] Marino, E., Yáñez, L., Guijarro, M., Madrigal, J., Senra, F., Rodríguez, S., Tomé, J.L., 2024. Transferability of empirical models derived from satellite imagery for live fuel moisture content estimation and fire risk prediction. *Fire* 7, 276. doi:10.3390/fire7080276.
- 1985 [23] Meyer, H., Pebesma, E., 2021. Predicting into unknown space? Estimating the area of applicability of spatial prediction models. *Methods in Ecology and Evolution* 12, 1620–1633. doi:10.1111/2041-210X.13650.
- [24] Meyer, H., Pebesma, E., 2022. Machine learning-based global maps of ecological variables and the challenge of assessing them. *Nature Communications* 13, 2208. doi:10.1038/s41467-022-29838-9.
- 1990 [25] Meyer, H., Reudenbach, C., Hengl, T., Katurji, M., Nauss, T., 2018. Improving performance of spatio-temporal machine learning models using

- forward feature selection and target-oriented validation. *Environmental Modelling & Software* 101, 1–9. doi:10.1016/j.envsoft.2017.12.001.
- [26] Milà, C., Mateu, J., Pebesma, E., Meyer, H., 2022. Nearest neighbour distance matching leave-one-out cross-validation for map validation. *Methods in Ecology and Evolution* 13, 1304–1316. doi:10.1111/2041-210X.13851.
- [27] Moreno-Torres, J.G., Raeder, T., Alaiz-Rodríguez, R., Chawla, N.V., Herrera, F., 2012. A unifying view on dataset shift in classification. *Pattern Recognition* 45, 521–530. doi:10.1016/j.patcog.2011.06.019.
- [28] Muñoz-Sabater, J., Dutra, E., Agustí-Panareda, A., Albergel, C., Arduini, G., Balsamo, G., Boussetta, S., Choulga, M., Harrigan, S., Hersbach, H., Martens, B., Miralles, D.G., Piles, M., Rodríguez-Fernández, N.J., Zsoter, E., Buontempo, C., Thépaut, J.N., 2021. ERA5-Land: a state-of-the-art global reanalysis dataset for land applications. *Earth System Science Data* 13, 4349–4383. doi:10.5194/essd-13-4349-2021.
- [29] Pausas, J.G., Keeley, J.E., 2021. Wildfires and global change. *Frontiers in Ecology and the Environment* 19, 387–395. doi:10.1002/fee.2359.
- [30] Persello, C., Bruzzone, L., 2012. Active learning for domain adaptation in the supervised classification of remote sensing images. *IEEE Transactions on Geoscience and Remote Sensing* 50, 4468–4483. doi:10.1109/TGRS.2012.2192740.
- [31] Ploton, P., Mortier, F., Réjou-Méchain, M., Barbier, N., Picard, N., Rossi, V., Dormann, C., Cornu, G., Viennois, G., Bayol, N., Lyapustin,

- 2015 A., Gourlet-Fleury, S., Pélissier, R., 2020. Spatial validation reveals poor predictive performance of large-scale ecological mapping models. *Nature Communications* 11, 4540. doi:10.1038/s41467-020-18321-y.
- [32] Podschwit, H., Jolly, W., Alvarado, E., Verma, S., Ponce, B., Markos, A., Aliaga-Nestares, V., Rodriguez-Zimmermann, D., 2022. Reliability of cross-regional applications of global fire danger models: A Peruvian case study. *Fire Ecology* 18, 25. doi:10.1186/s42408-022-00150-7.
- 2020 [33] Roberts, D.R., Bahn, V., Ciuti, S., Boyce, M.S., Elith, J., Guillera-Arroita, G., Hauenstein, S., Lahoz-Monfort, J.J., Schröder, B., Thuiller, W., Warton, D.I., Wintle, B.A., Hartig, F., Dormann, C.F., 2017. Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography* 40, 913–929. doi:10.1111/ecog.02881.
- [34] Rousseau, J.S., Betts, M.G., 2022. Factors influencing transferability in species distribution models. *Ecography* 2022, e06060. doi:10.1111/ecog.06060.
- 2030 [35] Sandholt, I., Rasmussen, K., Andersen, J., 2002. A simple interpretation of the surface temperature/vegetation index space for assessment of surface moisture status. *Remote Sensing of Environment* 79, 213–224. doi:10.1016/S0034-4257(01)00274-7.
- [36] Schoener, T.W., 1968. The *Anolis* lizards of Bimini: Resource partitioning in a complex fauna. *Ecology* 49, 704–726. doi:10.2307/1935534.
- 2035

- [37] Sun, B., Feng, J., Saenko, K., 2016. Return of frustratingly easy domain adaptation, in: Proceedings of the Thirtieth AAAI Conference on Artificial Intelligence (AAAI-16), AAAI Press, Phoenix, Arizona, USA. pp. 2058–2065. doi:10.1609/aaai.v30i1.10306.
- 2040 [38] Tuia, D., Persello, C., Bruzzone, L., 2016. Domain adaptation for the classification of remote sensing data: An overview of recent advances. IEEE Geoscience and Remote Sensing Magazine 4, 41–57. doi:10.1109/MGRS.2016.2548504.
- [39] Valavi, R., Elith, J., Lahoz-Monfort, J.J., Guillera-Arroita, G., 2019. blockCV: An R package for generating spatially or environmentally
2045 separated folds for k-fold cross-validation of species distribution models. Methods in Ecology and Evolution 10, 225–232. doi:10.1111/2041-210X.13107.
- [40] VanDerWal, J., Shoo, L.P., Graham, C., Williams, S.E., 2009. Selecting
2050 pseudo-absence data for presence-only distribution modeling: How far should you stray from what you know? Ecological Modelling 220, 589–594. doi:10.1016/j.ecolmodel.2008.11.010.
- [41] Vesk, P.A., Morris, W.K., Neal, W.C., Mokany, K., Pollock, L.J., 2021. Transferability of trait-based species distribution models. Ecography 44,
2055 134–147. doi:10.1111/ecog.05179.
- [42] Vibhandik, P., Sawant, S., Joshi, A., Bidwe, R., 2026. A systematic literature review on forest fire susceptibility mapping using geo-spatial

technology and future research directions. *Discover Artificial Intelligence* 6, 921. doi:10.1007/s44163-026-00921-0.

- 2060 [43] Wadoux, A.M.J.C., Heuvelink, G.B.M., de Bruin, S., Brus, D.J., 2021. Spatial cross-validation is not the right way to evaluate map accuracy. *Ecological Modelling* 457, 109692. doi:10.1016/j.ecolmodel.2021.109692.
- [44] Warren, D.L., Glor, R.E., Turelli, M., 2008. Environmental niche equiv-
2065 alency versus conservatism: Quantitative approaches to niche evolution. *Evolution* 62, 2868–2883. doi:10.1111/j.1558-5646.2008.00482.x.
- [45] Xu, Y., Dai, Y., Chang, L., Wang, Q., Dong, Y., 2026. Does
your wildfire prediction model actually work, or just score well?
arXiv preprint arXiv:2605.18911. doi:10.48550/arXiv.2605.18911,
2070 arXiv:2605.18911.
- [46] Yates, K.L., Bouchet, P.J., Caley, M.J., Mengersen, K., Randin, C.F.,
Parnell, S., et al., 2018. Outstanding challenges in the transferability of
ecological models. *Trends in Ecology & Evolution* 33, 790–802. doi:10.
1016/j.tree.2018.08.001.
- 2075 [47] Yebra, M., Dennison, P.E., Chuvieco, E., Riaño, D., Zylstra, P., Hunt,
E.R., Danson, F.M., Qi, Y., Jurdao, S., 2013. A global review of remote
sensing of live fuel moisture content for fire danger assessment: Moving
towards operational products. *Remote Sensing of Environment* 136,
455–468. doi:10.1016/j.rse.2013.05.029.

- 2080 [48] Zanaga, D., Van De Kerchove, R., Daems, D., De Keersmaecker, W., Brockmann, C., Kirches, G., Wevers, J., Cartus, O., Santoro, M., Fritz, S., Lesiv, M., Herold, M., Tsendbazar, N.E., Xu, P., Ramoino, F., Arino, O., 2022. ESA WorldCover 10 m 2021 v200. Zenodo. doi:10.5281/zenodo.7254221.